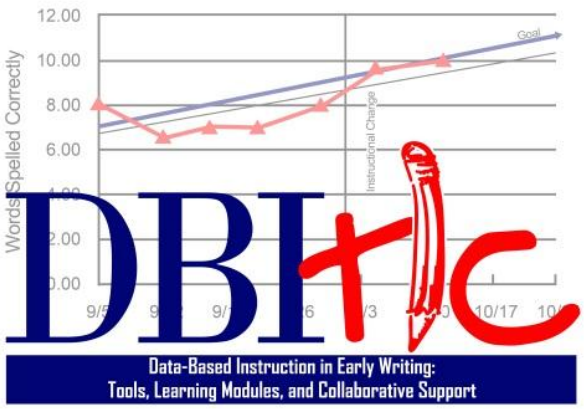


Data-Based Instruction in Beginning Writing: A Manual

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Adapted from the *Data-Based Program Modification Manual* (Deno & Mirkin, 1977), *Using CBM for Progress Monitoring in Written Expression and Spelling* (L.S. Fuchs & D. Fuchs, 2007), and materials from the National Center on Intensive Intervention (NCII; http://www.intensiveintervention.org/).

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Forward

Many students respond positively to research-based interventions that are implemented with fidelity -- by following the intervention protocol according to methods demonstrated to be effective. Yet, a small number of students do not benefit sufficiently from these interventions. These students require more intensive instruction designed to meet their individual instructional needs. Data-Based Instruction (DBI) provides a framework for teachers to individualize instruction systematically and effectively. Researchers have demonstrated that, when teachers implement DBI with fidelity, their instructional decision-making improves, and student achievement increases (see Stecker, Fuchs, & Fuchs, 2005 for a review).

Educators use data-based decision-making formally and informally each day in schools. However, educators do not always implement formalized DBI models consistently across content areas. With heightened pressure on schools and teachers to not only use data to make decisions but also to make decisions on a frequent basis and use the best evidence available to support their practices, it becomes increasingly necessary to incorporate a strong model to guide decision-making.

The purpose of this manual is to provide basic steps in the process of DBI in beginning writing. We have modeled the manual after similar work conducted by Deno & Mirkin (1977), L.S. Fuchs & D. Fuchs (2007), and materials from the National Center on Intensive Intervention (NCII; http://www.intensiveintervention.org/). The manual is intended to be used in conjunction with high quality, research-based intervention strategies in beginning writing. The information contained in this manual can be used for training purposes, as a resource for schools or individual teachers, as a guide to assist problem-solving teams or Response to Intervention teams in their decision-making, and as a model for special educators to intensify instruction. We have solicited feedback from teachers, administrators, and experts in the field to not only create the manual, but also make it user-friendly. We hope we have succeeded and wish you success as you use this manual!

How to Use this Manual

This manual is designed to be used as part of ***DBI-TLC***, a professional development system that includes:

> **T**ools for assessment, intervention, and data-based decision-making within the DBI framework, **L**earning modules that are provided via face-to-face workshops and/or web-based tutorials, and **C**ollaborative support provided on an ongoing basis to DBI implementers.

Throughout this manual, we refer you to each of the three **DBI Toolkits**, which contain the procedures and materials needed to implement DBI successfully.

DBI TOOLKITS:



Curriculum-Based Measurement Toolkit

- ☐ Selecting an Appropriate CBM Task
- ☐ CBM-W Prompts
- ☐ Administration Instructions
- ☐ Scoring Procedures
- ☐ Graphing Tools
- ☐ CBM Benchmarks
- ☐ Fidelity of CBM Administration



Writing Intervention Toolkit

- ☐ Instructional Alignment Tool
- ☐ Diagnostic Checklist (and other Diagnostic Tools)
- ☐ Writing Instructional Plans (WIP) (Blank Template and Transcription Example)
- ☐ Skill-Based Mini-Lessons
- ☐ Fidelity of Intervention Implementation



Decision-Making Toolkit

- ☐ Decision-Making Rubric
- ☐ Decision Log
- ☐ Fidelity of Data-Based Decision Making

We encourage you to take advantage of the illustrations, practice, and application activities, which were designed to help you learn to implement DBI effectively and efficiently to improve students' beginning writing outcomes.

Manual Objectives

In the next sections, you will learn how to apply the DBI steps to individualize instruction for beginning writers. Throughout the manual, we will refer to a case example to help you envision the DBI process with an actual student. After reading this manual, you will be able to:

1. Describe DBI for beginning writers, including what it is, why to implement it, who should receive it, and the components of writing that may be addressed with DBI.
2. Describe Curriculum-Based Measurement (CBM) tasks for beginning writers.
3. Select, administer, score, and graph data from CBM for beginning writers.
4. Implement the eight steps of DBI with fidelity.

OBJECTIVE 1: Describe data-based instruction for beginning writers.

What is DBI for Beginning Writers?

DBI is a systematic, data-based approach for teachers to individualize instruction for beginning writers, using the steps shown in Figure 1.

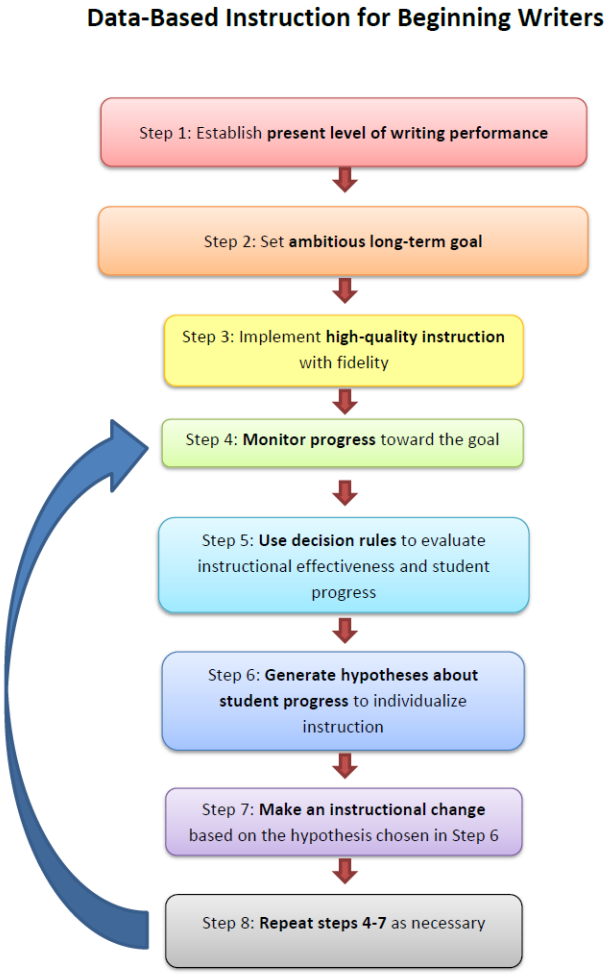


Figure 1. ***Eight Steps in the DBI Process.***

Why Implement DBI?

Many students respond positively to research-based interventions that are implemented with ***fidelity*** –by following the intervention protocol according to methods demonstrated to be effective. Yet, a small number of students do not benefit sufficiently from these interventions. These students require more intensive instruction designed to meet their individual instructional needs. DBI provides a framework for teachers to individualize instruction systematically and effectively. Researchers have demonstrated that, when teachers use DBI with fidelity, their instructional decision-making improves, and student achievement increases (see Stecker, Fuchs, & Fuchs, 2005 for a review).

Who Should Receive DBI?

Currently, many schools are implementing Multi-Tiered Systems of Support (MTSS), such as Response to Intervention (RTI). Within such models, students for whom core instruction (e.g., primary prevention or Tier 1) and/or supplementary intervention (e.g., secondary prevention or Tier 2) have not been sufficient receive more individualized, intensive instruction (e.g., tertiary intervention or Tier 3). In some schools and districts, Tier 3 intervention precedes special education referral; in others, Tier 3 *is* special education. In either case, DBI is intended for those students (including students in special education) who require intensive, individualized instruction beyond that provided in core instruction or supplementary intervention.

DBI Assumptions

Data-Based Instruction (DBI) is based on the following assumptions (adapted from Deno & Mirkin, 1977; Research Institute on Problem Solving; www.cehd.umn.edu/EdPsych/rips/).

- To provide effective individualized instruction, educators should implement instructional approaches that are ***research based.***
- Effective, research-based instructional approaches exist, but it is impossible to predict whether these approaches will meet the unique needs of each individual student. Educators can only hypothesize that a given instructional approach will work for an individual student; thus, educators must test whether it is effective for that student***.***
- Educators can collect ongoing ***assessment data*** and use it as ***evidence*** to determine whether an instructional approach is working for an individual student.

The ongoing assessment data used for instructional decision-making should reflect ***critical academic skills*** that we expect to ***improve over time.***

Components of writing addressed in DBI.

DBI requires that we know which academic skills to target for assessment and intervention. For beginning writers, we use a framework called the "Simple View of Writing" developed by Virginia Berninger and her colleagues (e.g., Berninger & Amtmann, 2003). The Simple View of Writing, described below and illustrated in Figure 2, specifies three major components of the writing process:

- ***Text generation*** means turning ideas into text at **three levels of language: *words***, ***sentences***, and ***passages***. You can think of text generation as coming up with ideas (which relate to a specific genre, text structure, and topic), and then coming up with the right words, sentences, or whole passages to express those ideas.
- ***Transcription*** means translating sounds, words, sentences, and passages (the text that the writer *generates)* into print. Transcription includes handwriting, spelling, and mechanics (punctuation, etc.)
- ***Self-regulation*** is what writers do to meet their writing goals. Self-regulation includes goal setting, planning, organizing, self-monitoring, self-evaluating, revising, and self-rewarding, among other approaches to controlling and managing the writing process.

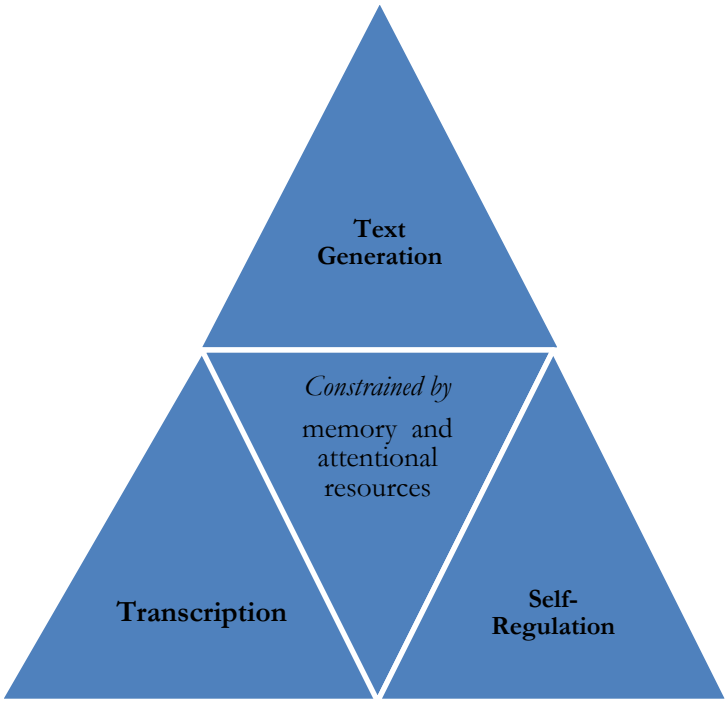


Figure 2. ***Components of Writing.***

Researchers (e.g., Berninger, 2009; Graham et al., 2012; McCutchen, 2006) have shown that each of these components is important to the overall quality of students' writing. When students struggle with basic skills (such as handwriting and spelling), they have limited memory and attention to devote to text generation and self-regulation. These three components can be used to guide the focus of assessment and intervention for beginning writers.

Applied Example: Mrs. Lewis and her student, Molly

So what does DBI look like in a classroom? Let's take the case of a special education teacher who wants to implement DBI. How would she start? What materials would she need? We illustrate the basic DBI steps throughout the remainder of the manual using a case example of Mrs. Lewis, a special education teacher who is working with a student named Molly.

Molly is a third-grade student who receives special education services focused on written language, including reading and written expression. She was identified for special education at the end of her second-grade year. Her IEP team determined she would receive daily small-group instruction in reading and writing by the special education teacher, Mrs. Lewis. Mrs. Lewis knows that she will need to track Molly's progress on a frequent basis to make sure that the instruction is working for Molly. Mrs. Lewis wants to use some type of assessment process that is easy to implement, *and* gives her good information about how Molly is doing overall in writing. This is how Mrs. Lewis's thinking leads her to the DBI process.

To get started, Mrs. Lewis needs more information about Molly and her present performance level. Mrs. Lewis uses the *Selecting an Appropriate CBM Task* tool (in CBM Toolkit) to select a measure that will help establish Molly's present level of writing performance. A completed copy of this document for Molly is in Appendix A.

Table 1 provides a description of how Mrs. Lewis implemented the eight steps of the DBI process, including a sample graph (Figure 3), to help you conceptualize how the DBI process works. The eight basic steps in the DBI process are explained in detail throughout the rest of the manual.

Table 1. Eight Steps of DBI

DBI Step *(Click hyperlinked text [in blue] in each step for more information.)*	**Found In:**
1. Establish present level of writing performance (baseline). Using the *Selecting an Appropriate CBM Task* tool (in CBM Toolkit; see Molly's example in Appendix A), Mrs. Lewis chooses to administer Picture Word Prompts and administers three prompts. She scores each prompt using Correct Word Sequences (CWS), a standard CBM scoring method. On the three prompts, Molly obtains 5, 7 and 4 Correct Word Sequences. Mrs. Lewis plots the median, 5 CWS, on the graph as Molly's BASELINE (red diamond on the graph).	Objective 3 Objective 4, Step 1; Appendix A; CBM Toolkit
2. Set an ambitious long-term goal. Mrs. Lewis wants to determine a reasonable, but ambitious, goal for Molly to grow in writing over the school year. She determines a goal based on the school's third grade benchmark for writing in spring, and plots this LONG-TERM GOAL on the graph (blue star) at the end of the school year. She draws a GOAL LINE from the BASELINE to the LONG-TERM GOAL.	Objective 4, Step 2; CBM Toolkit
3. Implement high-quality instruction based on student needs. Mrs. Lewis draws a vertical line after the baseline, and labels the new section INTERVENTION 1. She consults the *Instructional Alignment Tool* and the *Diagnostic Checklists* (in Writing Intervention Toolkit) and decides to focus on handwriting, spelling, and sentence construction—skills determined by Molly's IEP team to be critical for Molly's success in school. She uses these skills to design a *Writing Instructional Plan* (in Writing Intervention Toolkit; see Molly's example in Appendix A).	Objective 4, Step 3; Appendix A; Writing Intervention Toolkit

Table 1 *continued*

DBI Step *(Click hyperlinked text [in blue] in each step for more information.)*	**Found In:**
4. Monitor progress toward the goal using CBM. Mrs. Lewis monitors Molly's progress toward the goal by giving a 3-minute CBM Picture-Word prompt each week. She continues to score each prompt using CWS. She plots Molly's progress monitoring score each week in the INTERVENTION 1 section on the graph. After collecting and plotting multiple data points (best practice suggests 8 data points or weekly progress monitoring scores), Mrs. Lewis adds a trend line.	Objective 4, Step 4; CBM Toolkit
5. Use decision rules to evaluate student progress and instructional effectiveness. A "decision rule" is a rule that tells when to make an instructional change. Mrs. Lewis uses a decision rule that says that if the trend line is less steep than the goal line, she should CHANGE INSTRUCTION. Mrs. Lewis sees that Molly's progress falls below the goal line. So, she decides to make an instructional change.	Objective 4, Step 5
6. Generate hypotheses about student progress to individualize instruction. Based on her examination of Molly's progress monitoring data collected during intervention, and using the *Decision-Making Rubric* (in Decision-Making Toolkit and Appendix A) as a guide, Mrs. Lewis generates several hypotheses about why Molly is making insufficient progress, but then chooses the one that suggests that Molly needs more time in her current intervention.	Objective 4, Step 6; Decision-Making Toolkit
7. Make an instructional change based on the hypothesis generated in Step 6. Mrs. Lewis increases the frequency of intervention from 3 to 5 days per week. She draws a vertical line after INTERVENTION 1 and labels the new section INTERVENTION 2. She records the change on her *Decision Log* (in Decision-Making Toolkit; see Molly's example in Appendix A).	Objective 4, Step 7; Appendix A; Decision-Making Toolkit
8. Repeat steps 4 through 7. After collecting several weekly data points in Intervention 2, it is clear that increasing the frequency of instruction is not sufficient to improve Molly's progress. Mrs. Lewis continues to use the *Decision-Making Rubric* (in Decision-Making Toolkit) and generates a new hypothesis—that Molly needs more explicit instruction with additional modeling and practice. She adds this change to her instruction, and continues to monitor Molly's progress during INTERVENTION 3. During Intervention 3, Molly's progress increases; however, her performance is still below the goal line. Thus, Mrs. Lewis develops a new hypothesis—that Molly needs additional reinforcement for working on her writing, and would benefit from monitoring and charting her own progress toward her goal. She adds this change to her instruction, and continues to monitor Molly's progress during INTERVENTION 4. Finally, Molly meets and exceeds her goal. Now Mrs. Lewis can re-visit the *Instructional Alignment Tool* and *Diagnostic Checklists* (in Decision-Making Toolkit) to establish her next area of focus for Molly.	Objective 4, Step 8; Decision-Making Toolkit

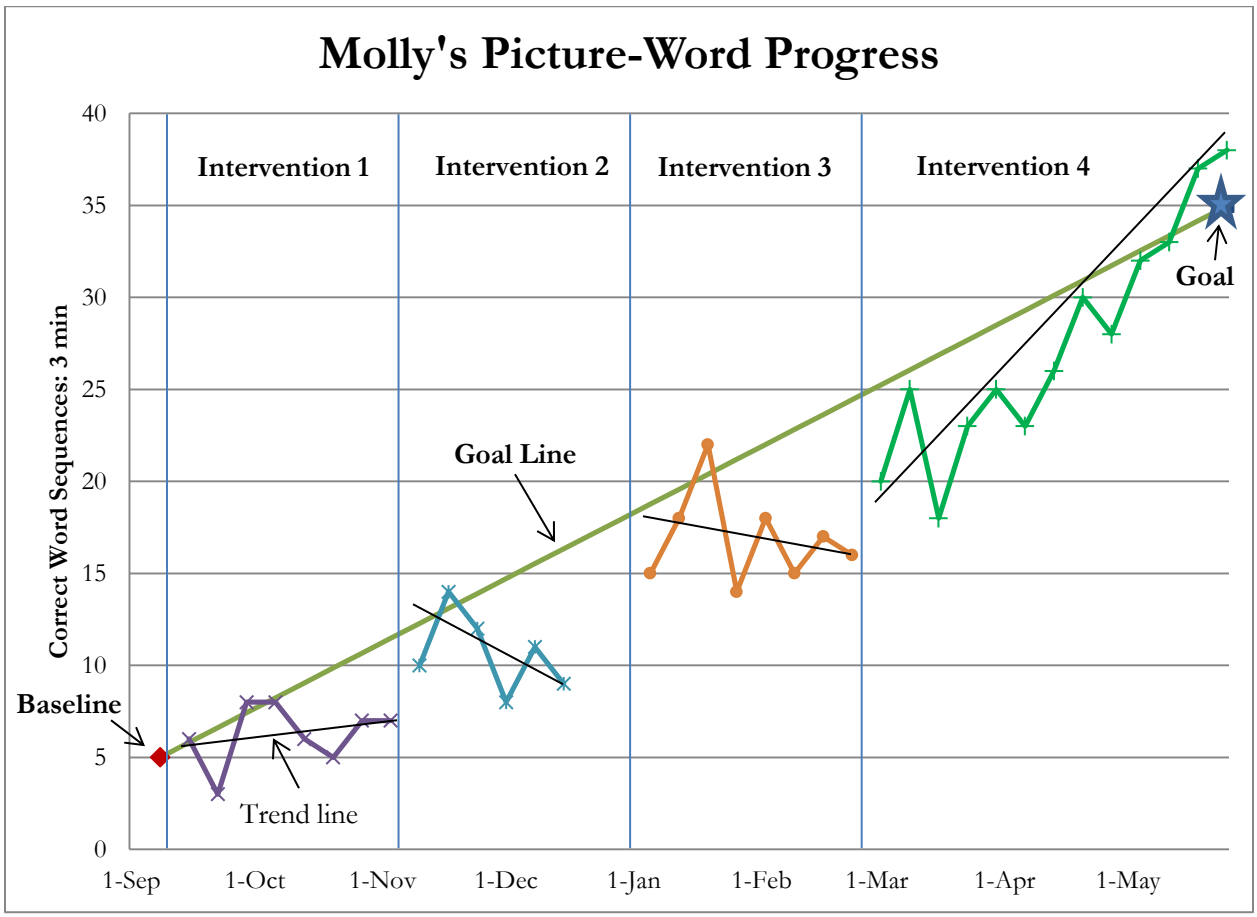


Figure 3. ***Monitoring Molly's Progress using the Eight Steps of DBI.***

CONSIDER THIS:

What components of writing does Mrs. Lewis target in her instruction with Molly?

Mrs. Lewis addresses **all three writing components** in her instruction with Molly. She implements a research-based intervention that focuses on **transcription** (hand-writing and spelling) and **text generation** (sentence construction). When Molly's CBM data indicate that this intervention is not sufficient, Mrs. Lewis provides more explicit instruction in these areas, which leads to an increase in Molly's progress. Molly's progress is further boosted when Mrs. Lewis adds a **self-regulation** component to the intervention, by having Molly monitor and chart her own progress. The Writing Intervention Toolkit will provide you with resources for intervention ideas in these areas.

OBJECTIVE 2: Describe CBM for beginning writers.

What is Curriculum-Based Measurement (CBM)?

A key assumption of DBI is that ongoing progress data used for instructional decision- making should reflect critical academic skills that we expect to improve over time (Deno & Mirkin, 1977). CBM is uniquely suited to DBI: It entails simple, efficient procedures that provide *global indicators* of student performance and progress in core academic domains. There are four key characteristics of CBM (Deno, 1985). They are:

1. ***Reliable and valid*** (the results are consistent and reflect the critical academic skills they are intended to measure).
2. ***Simple and efficient*** to administer and score, so that they may be used on a frequent basis.
3. ***Easy to understand*** so that the results can be clearly communicated to other school personnel, parents, and students.
4. ***Inexpensive*** given that multiple forms are needed for ongoing progress monitoring.

For More Information
on CBM in reading, writing, and mathematics, see Foegen, Jiban, & Deno, 2007; McMaster & Espin, 2007; Wayman, Wallace, Wiley, Ticha, & Espin, 2007 for reviews.

How is CBM different from other assessment approaches?

Perhaps you currently assess your student's writing performance using an informal writing rubric (such as the 6+1 Traits® Writing Rubric, Common Core State Standards). Or, you might assess your student's writing annually by administering a standardized test of writing skills. ***So, why would you need to use CBM?***

CBM in writing can supplement informal diagnostic writing rubrics and standardized writing assessments by providing different information. CBM can be given frequently (weekly or daily), is relatively easy to administer and score, and can measure very small increments of student progress. CBM provides a quick "temperature check" of student performance. It is not meant to replace other more comprehensive diagnostic assessments of writing performance. In this way, CBM serves as a *global indicator* of a student's writing "health." When CBM data are on track to meet important benchmarks or other goals, then the teacher can continue instruction as-is. But, when CBM data indicate that a student is not making sufficient progress, it signals the need for further diagnosis and intervention.



How do teachers make time for CBM?

Collecting data may seem time-consuming, but in the end, it can SAVE time and effort, and more importantly, prevent time from being wasted on instruction that is not effective. Teachers have made time for CBM in the following ways:

- Administer CBM to 2-3 students each day of the week.
- Administer CBM during small group time—keep one child at the end.
- Plan CBM into your regular schedule—Consider it part of instruction, because it is!
- Enter CBM data as you go—that is more efficient than letting it pile up!
- Find ways to hold yourself accountable (e.g., have periodic check-ins with principal, colleagues, parents, etc.)

CBM for Beginning Writers

CBM for beginning writers includes three tasks as part of text generation: word dictation, picture-word prompts, and story prompts. Figure 4 shows how each CBM task aligns with the writing components (transcription, text generations, and self-regulation) at three levels of language (word, sentence, and passage).

Each CBM writing task is described in this section. Research supporting each task is in Appendix B. Administration, scoring, and graphing procedures are described in the next section, under Objective 3. Prompts for each task are available in the CBM Toolkit.

CONSIDER THIS:

What *components of writing* and *level of language* does Mrs. Lewis target in monitoring Molly's progress?

Following completion of the *Selecting an Appropriate CBM Task tool* (in CBM Toolkit), Mrs. Lewis determines that Molly has difficulties in the areas of text generation and transcription at the sentence level. These difficulties correspond most closely with the Picture-Word Prompts.

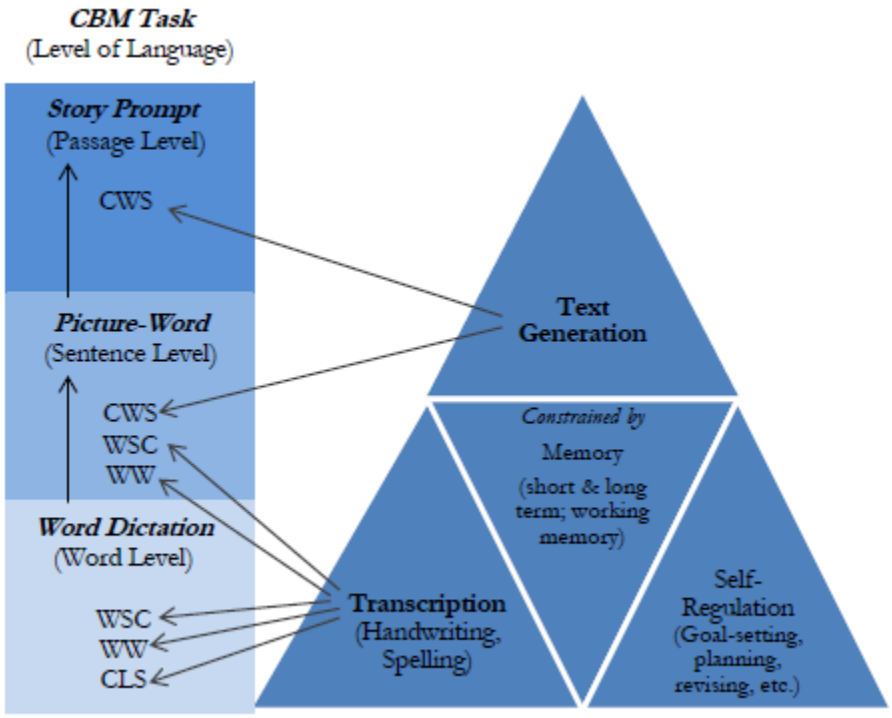


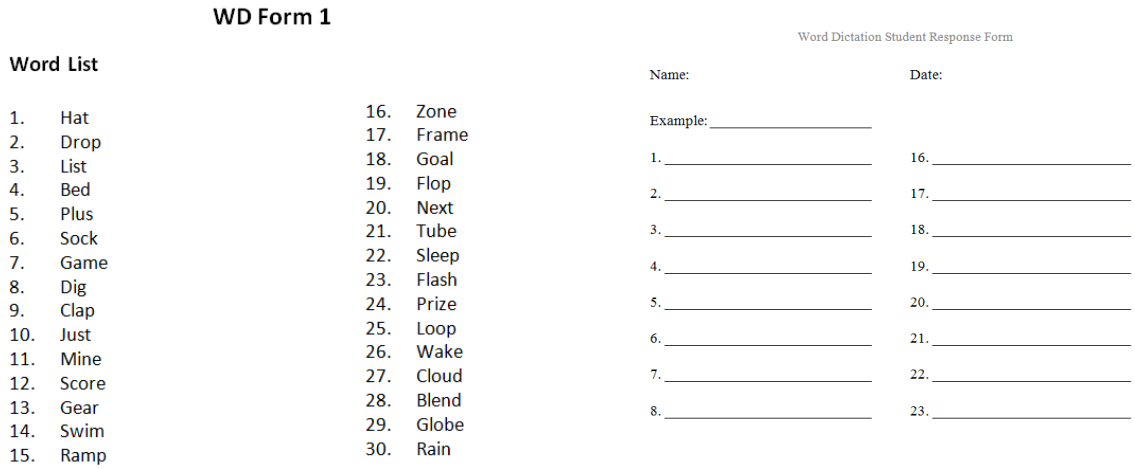
Figure 4. ***CBM Tasks Aligned with Levels of Language and Components of Writing.***
Note: CLS = Correct Letter Sequences, WW = Words Written, WSC = Words Spelled Correctly, CWS = Correct Word Sequences.

Word Dictation Prompts (word level).

Word Dictation Prompts are designed to capture ***transcription*** skills at the ***word level***. Word Dictation is appropriate for students who are just beginning to write words. Word Dictation Prompts were constructed by randomly sampling 6 types of word patterns (addressed in the Common Core State Standards for Grades 1-2), along with irregular words that young students frequently use in their writing (Graham & Harris, 2003). The six word patterns include:

- Consonant-Vowel-Consonant (CVC) words (e.g., cat)
- Consonant-Consonant-Vowel-Consonant (CCVC) words (e.g., slip)
- Consonant-Vowel-Consonant-silent e (CVCe) words (e.g., bake)
- Consonant-Consonant-Vowel-Consonant-silent e (CCVCe) words (e.g., plate)
- Consonant-Consonant-Vowel-Consonant-Consonant(CCVCC) words (e.g., plant)
- Consonant-Vowel-Vowel-Consonant (CVVC) words (e.g., soap)

Each Word Dictation Prompt contains 30 words and is administered individually for 3 minutes. Figure 5 shows a sample Word Dictation word list and student form.



WD Form 1

Word List

1. Hat
2. Drop
3. List
4. Bed
5. Plus
6. Sock
7. Game
8. Dig
9. Clap
10. Just
11. Mine
12. Score
13. Gear
14. Swim
15. Ramp
16. Zone
17. Frame
18. Goal
19. Flop
20. Next
21. Tube
22. Sleep
23. Flash
24. Prize
25. Loop
26. Wake
27. Cloud
28. Blend
29. Globe
30. Rain

Word Dictation Student Response Form

Name: Date:

Example: _____

1. _____ 16. _____
2. _____ 17. _____
3. _____ 18. _____
4. _____ 19. _____
5. _____ 20. _____
6. _____ 21. _____
7. _____ 22. _____
8. _____ 23. _____

Figure 5. ***Sample Word Dictation Word List and Student Form.***

Picture-Word Prompts (sentence level).
Picture-Word Prompts are designed to capture ***transcription*** and ***text generation*** at the ***sentence level***. Picture-Word Prompts are appropriate for students who are learning to connect words into sentences. Each prompt consists of multiple items. Each item consists of a picture as well as a word that is related to the picture. At the beginning of the task, the tester reads the word aloud to the student, while the student looks at the picture.

Each Picture-Word Prompt contains 12 pictures/words and is group-administered for 3 minutes. Figure 6 shows a sample page from a Picture-Word Prompt.

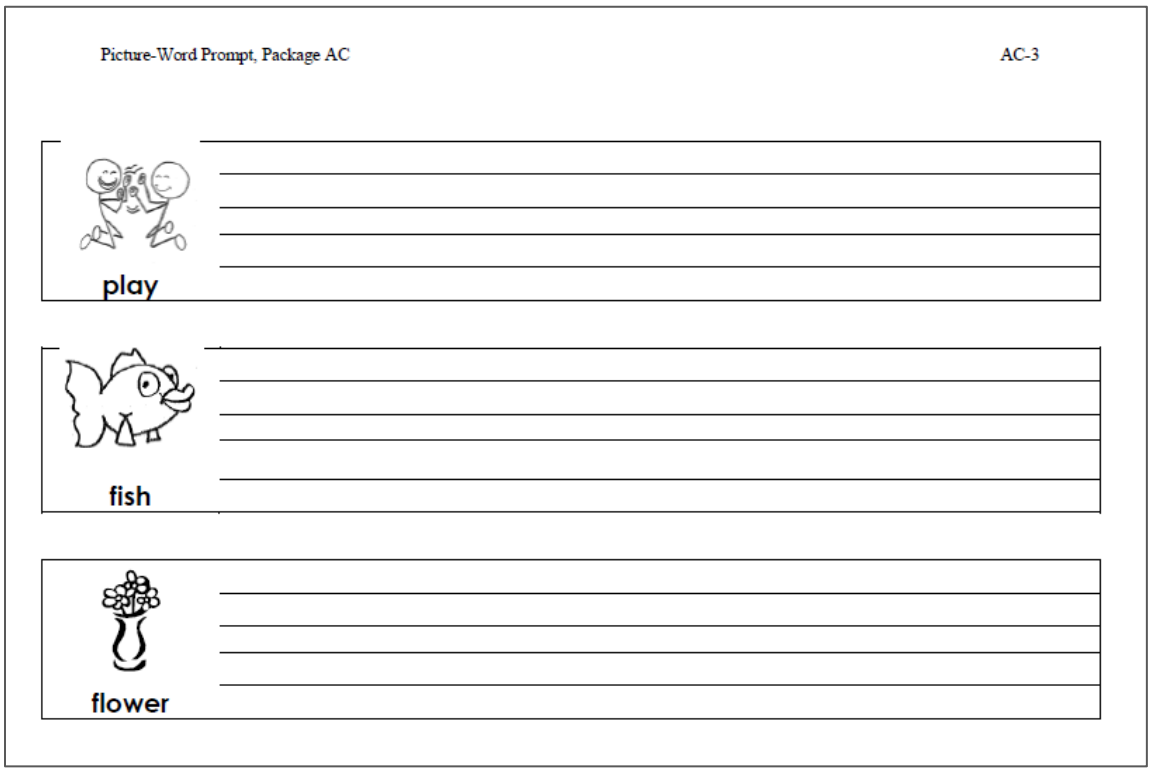


Figure 6. ***Sample page from a Picture-Word Prompt.***

Story Prompts (passage level).
Story Prompts are designed to capture ***transcription*** and ***text generation***. This task is appropriate for students who can generate ***passages.*** Story Prompts are simple, open-ended sentences, designed to tap age-appropriate background knowledge. They are designed to represent the experiences of a wide range of school-age students. Students generate a story in response to the Story (open-ended sentence) Prompt. Each prompt is group-administered for 3 minutes.

Figure 7 shows a sample page from a Story prompt.

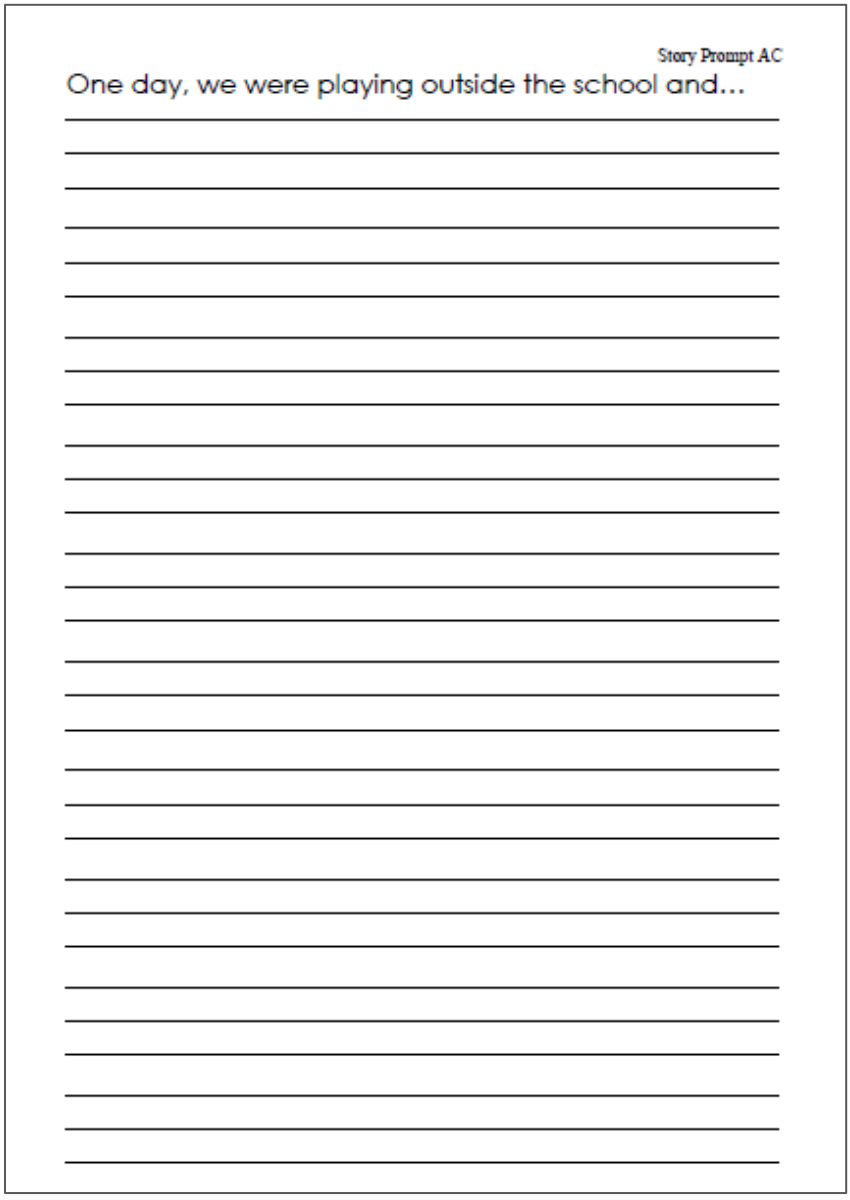


Figure 7. ***Sample Story Prompt.***

OBJECTIVE 3: Select, administer, score, & graph CBM for beginning writers.

1. Select an appropriate CBM progress monitoring task.

An important decision for implementing CBM is to determine which task to use to monitor student progress over the academic year. The first step is to benchmark the student's skill level by administering all three benchmark writing prompts (Word Dictation, Picture-Word Prompt, and Story Prompt) and to analyze the results.

Table 2 provides guidance about the level of language and approximate grade level for which each task has been designed and validated.

Table 2. ***Tasks by Level of Language and Approximate Grade Level, and Scoring Method****

Task	Level of Language	Grade Level	Scoring Method
Word Dictation	Word	1, 2, 3	Correct Letter Sequences (CLS), Words Written (WW), Words Spelled Correctly (WSC)
Picture-Word	Sentence	1, 2, 3	WW, WSC, Correct Word Sequences (CWS)
Story Prompt	Passage	2, 3	WW, WSC, CWS

To determine the appropriate type of CBM task to use, consult the *Selecting an Appropriate CBM task* (in CBM Toolkit) to consider the following:

- **The student's present skills**: Can the student produce whole words? Sentences? Longer passages?
- **The student's current instructional needs**: Does the student need to develop word-writing skills? Sentence-writing skills? Passage-writing skills?
- **The student's current grade level**.

CONSIDER THIS:

Which CBM task did Mrs. Lewis use for Molly? Why do you think she chose this task?

Molly is in third grade, but still needs to develop sentence-level writing skills. Given that she scores 5 CWS on Picture-Word Prompts, she may even need to develop word-level writing skills. Mrs. Lewis may have chosen Picture-Word Prompts because it is likely to be more sensitive to Molly's progress than Story Prompts, but more in line with grade-level skills than Word Dictation.

2. Administer the CBM task using the standardized set of instructions.

CBM tasks must be administered the SAME WAY each time, so that you can monitor student progress in a meaningful way. The following tips will help ensure accurate administration of CBM. ***Specific administration instructions for each task are provided in the CBM Toolkit.***

- **Be sure to have all materials needed prior to administration:**
  - Timer
  - Sharpened pencils
  - Directions for administration
  - Teacher copy of the task to demonstrate (or put on smart board)
  - Copy of task for students
- **Administer each CBM task according to standardized administration directions for the prescribed length of time.** This way, when the student's performance improves or decreases, you'll know the change in performance is due to changes in the student's writing skills, not due to changes in the directions you're using to administer the CBM task.
- **Administer CBM under standardized conditions**, including the same day(s) of the week, time of day, location, and administrator. Changing any of these conditions might affect student performance, leading to scores that might be difficult to interpret.
- **Consider the student and the environment:**
  - Make sure the student is seated comfortably, has a pencil he/she is used to using for writing, and has enough space to write.
  - Minimize visually distracting materials on the wall/table where you are testing
  - Avoid times when disruptions are likely (morning announcements, Valentine's day party, loud classrooms going by, assemblies)

3. Score the CBM task in a timely manner.

After each CBM administration, it is important to score the student's writing sample as soon as possible. Scoring in a timely manner will make the DBI process much more manageable and effective. Scoring methods for each task are listed in Table 2 above, and described below. ***Specific scoring procedures and practice items are provided in the CBM Toolkit.***

- **Words Written (WW):** The total number of words written. A "word" is defined as a sequence of letters separated by a space from another sequence of letters.
- **Words Spelled Correctly (WSC):** The number of correctly spelled words written in the sample (see Scoring Procedures for more detail).
- **Correct Letter Sequences (CLS):** Any two adjacent letters that are correct according to the spelling of the word.
- **Incorrect Letter Sequences (ILS):** Any two adjacent letters that are incorrect according to the spelling of the word.
- **Correct Word Sequences (CWS):** Any two adjacent words that are correct in terms of spelling, grammar, capitalization, and punctuation in the context of the sentence.
- **Incorrect Word Sequences (IWS):** Any two adjacent words that are incorrect in terms of spelling, grammar, capitalization, and punctuation in the context of the sentence.

4. Establish inter-rater agreement.

It is important that you and your colleagues are accurate and consistent in how you score students' CBM samples. That way, you'll be sure that the students' scores are consistent with the standardized scoring procedures.

We suggest that you and your colleagues score the same samples independently (at least 1 out of every 10 of each type of CBM task administered) and compare your scores. If there are discrepancies in your scores, review the directions carefully to "recalibrate" your scoring so it matches the standardized scoring directions. Then try again with a different set of samples. This is called **establishing inter-rater agreement.** Practice samples for establishing inter-rater agreement for each CBM task are provided in CBM Toolkit.

1. With your colleagues, read the scoring procedures carefully. Discuss any procedures that are unclear.
2. Practice scoring a student CBM sample together, discussing the rules as you go.
3. As needed, practice scoring additional samples until everyone is comfortable with the scoring procedures. If needed, make notes on the scoring procedures document to clarify ambiguous issues. The goal is for everyone to make scoring decisions in a consistent way.
4. When you have collected a number of student CBM samples (e.g., samples from an entire class), select 10% of those samples at random.
5. Make copies of the samples and have each teacher score the *same* samples, independently.
6. When everyone has scored the samples, establish one person as the primary scorer (this person should be the one who is likely to score the most CBM samples in the future).
7. Compare each person's scores to those of the primary scorer, examining each item (letter sequence, word written, word spelled, or word sequence). If the second scorer *agreed* with the primary scorer, it is counted as an "agreement." If the second scorer *disagreed* with the primary scorer, it is counted as a "disagreement."
8. Add up the number of agreements, and the number of agreements plus disagreements.
9. Divide the number of agreements by the number of agreements plus disagreements and multiply by 100:

$$\frac{\text{Agreements}}{\text{Agreements + Disagreements}} \times 100 = \text{\% Inter-Rater Agreement}$$

10. For each scorer, if agreement is at least 85%, you may begin scoring separate samples.
11. For scorers with agreement less than 85%, you should discuss and resolve areas of disagreement, then repeat steps 1-9 until at least 85% agreement is reached.

5. Graph student data on a regular basis.

Graphing student CBM data on a regular basis is a critical component of DBI. Graphed data will provide you with a clear picture of a student's progress. The graph will allow you to:

- Set reasonable and ambitious goals,
- Monitor the appropriateness of the student's goal,
- Judge the adequacy of the student's progress,
- Determine the effectiveness of the student's writing instructional program,
- Use decision rules to make changes to the student's instructional program when needed.

Your school district may have a data management system that allows you to graph progress monitoring data. If not, you can use computer software to create your own graphs. One of the most common types of software for chart making is Microsoft Excel. It can be a valuable tool for teachers, because once you have set up a graph, it is easy to update it whenever you collect additional scores. See tutorial for step-by-step instructions for entering data and creating graphs using Excel (in the CBM Toolkit).

OBJECTIVE 4: Implement the eight DBI steps with fidelity.

In this section, you will learn to implement each step in the DBI process (introduced at the beginning of this manual). For each step, we indicate which of the following Toolkits you will need:



Curriculum-Based Measurement Toolkit

- ☐ Selecting an Appropriate CBM Task
- ☐ CBM-W Prompts
- ☐ Administration Instructions
- ☐ Scoring Procedures
- ☐ Graphing Tools
- ☐ CBM Benchmarks
- ☐ Fidelity of CBM Administration



Writing Intervention Toolkit

- ☐ Instructional Alignment Tool
- ☐ Diagnostic Checklist (and other Diagnostic Tools)
- ☐ Writing Instructional Plans (WIP) (Blank Template and Transcription Example)
- ☐ Skill-Based Mini-Lessons
- ☐ Fidelity of Intervention Implementation



Decision-Making Toolkit

- ☐ Decision-Making Rubric
- ☐ Decision Log
- ☐ Fidelity of Data-Based Decision Making

Each step includes the following:



DESCRIPTION:
We include a rationale, description, and examples of the step.



ILLUSTRATION:
We use Molly's case example to illustrate how the step can be applied in practice.



PRACTICE:
We provide an opportunity for you to practice the step with a brief activity.



APPLICATION:
We encourage you to apply the step with an actual student.

Step 1: Establish present level of writing performance (baseline) using CBM.

Toolkits needed: CBM & DECISION-MAKING

DESCRIPTION: Establishing the student's present level of educational performance, or baseline, is a critical step for setting the student's long-range goal.

a. Select a CBM task (Word Dictation, Picture-Word, or Story Prompt) that you will use to monitor a student's progress using the *Selecting an Appropriate CBM Task tool* (in CBM Toolkit).
b. Administer three *different* prompts for that CBM task (the *task* should stay the same, but the *prompts* should be different). You can administer the prompts in one day or across 3 days—preferably all within one week.
c. Score the three baseline prompts, using the standardized scoring directions.
d. Determine the **median score** of the three prompts. To determine the median score, rank order the three scores from lowest to highest. The middle score is the median.
e. Graph this baseline score (see Molly's graph below, which shows a baseline of 5, plotted on the graph as a red diamond).

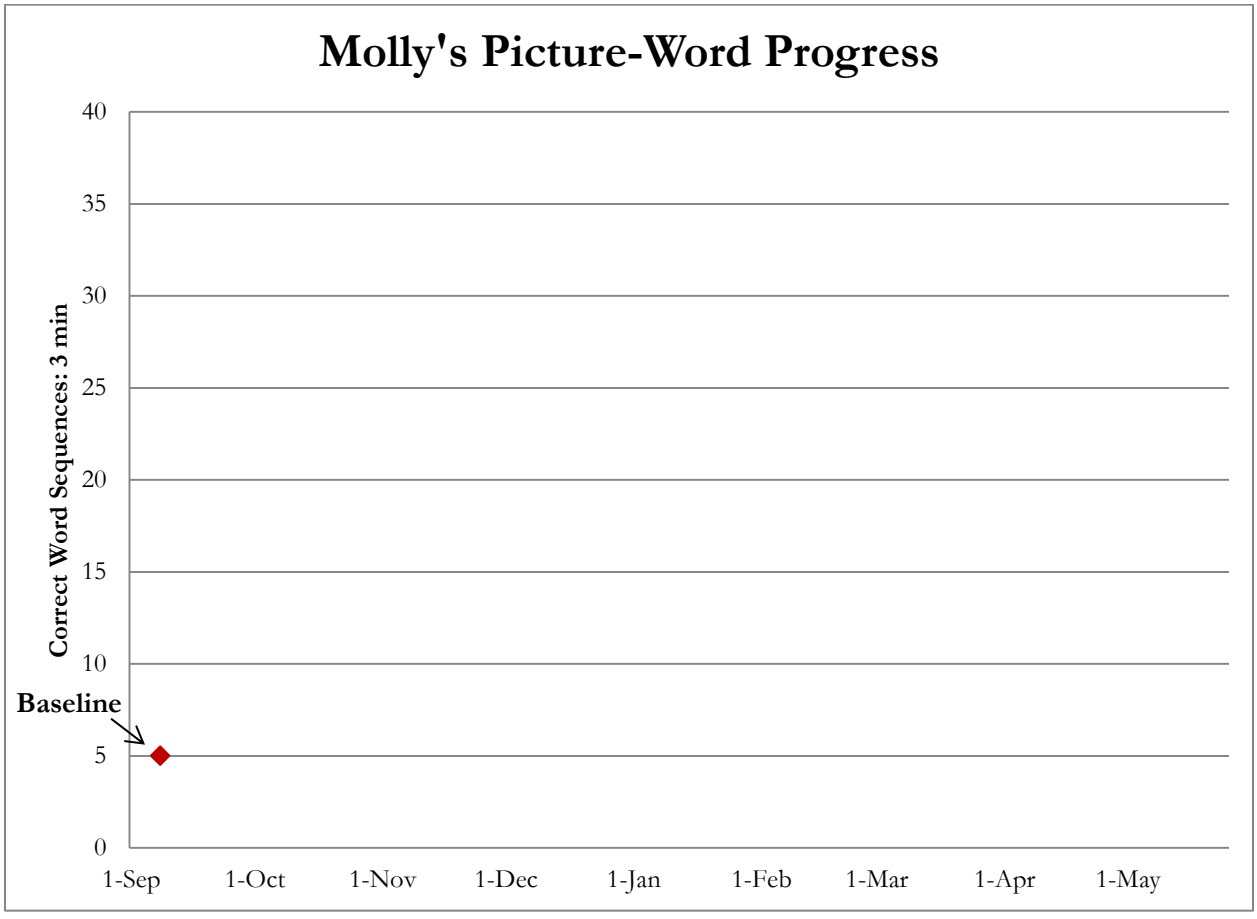




ILLUSTRATION: Mrs. Lewis uses the *Selecting an Appropriate CBM Task* (in CBM Toolkit), and chooses to administer Picture-Word Prompts and administers three prompts.

She scores each prompt using Correct Word Sequences (CWS), a standard CBM scoring method. On the three prompts, Molly obtains 5, 7 and 4 Correct Word Sequences. Mrs. Lewis plots the median, 5 CWS, on the graph as Molly's BASELINE (red diamond on the graph). [Click here to view entire case example.]



PRACTICE: Find the median of the following scores: 12, 15, 10

Plot this median on a graph. This will be your Practice Graph. (Practice Graph data and graph are in Appendix C.)



APPLICATION: Choose one student whose progress you wish to monitor. Administer three baseline writing samples using Word Dictation, Picture-Word, *or* Story Prompts. Choose a scoring procedure (CLS, WW, WSC, or CWS). Score each sample, and the median of the three scores. Graph the median, using Excel or your school program. This will be your Student Graph.

Step 2: Set an ambitious long-term goal.

Toolkit needed: CBM

DESCRIPTION: Setting ambitious goals ensures high expectations of students, which will lead to the best instruction possible. To set an ambitious long-term goal:

a. Determine the end-date of the instructional period in which you will monitor progress. The instructional period is often the end of the semester or school year.
b. Determine the level at which you expect the student to perform at the end of this instructional period (see Options 1-3 below). This level is the student's long-term goal.
c. Plot the long-term goal point on the last date of the instructional period. For example, in Molly's graph above, the long-term goal is 17 CWS, indicated by the blue star.
d. Draw a goal line by connecting the baseline and the long-term goal.

There are **three options** for determining the long-term goal: end-of-year benchmark, normative growth rate, and individualized student approach:

OPTION 1: End-of-Year Benchmark

Identify the end-of-year CBM benchmark for typically developing students at the grade level at which the student is being monitored. Use this benchmark as the long-term goal.

NOTE:

Benchmarking recommendations for Grades 1–3 are in the CBM Toolkit. However, we strongly recommend that the teacher, school, or district create local benchmarks (based on classroom, school, or district performance).

To set a goal for the student:

- Plot the student's **baseline** score on the graph on the first date of the instructional period in which you plan to monitor progress.
- Plot the **end-of-year benchmark** on the graph on the last date of the instructional period in which you plan to monitor progress.
- Draw a goal line from **baseline** to **end-of-year benchmark**.

If you are using a spreadsheet program like Excel, you will need to figure out the **desired growth rate** to get to the end-of-year benchmark:

- Subtract the **baseline** score from the **end-of-year benchmark**. This is the number the student needs to gain to reach the benchmark.
- Divide this number by the number of weeks remaining in the instructional period to get a **desired growth rate**.
- The goal line is created by increasing each expected weekly score by the **desired growth rate** (see example below).

END-OF-YEAR BENCHMARK EXAMPLE

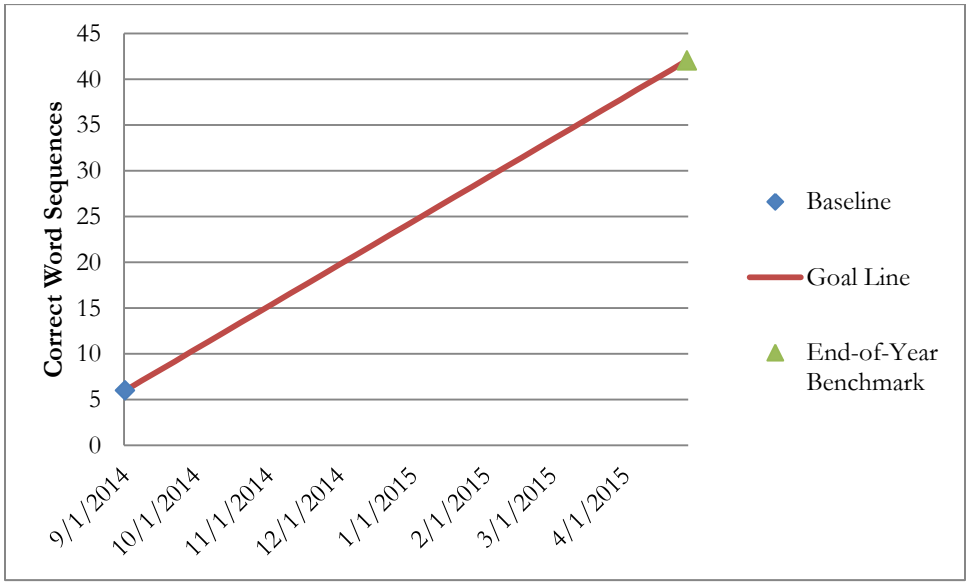
A second-grade student's median score from her first three Picture-Word Prompts is 6 CWS. The district end-of-year benchmark for second-graders is 42 CWS. There are 34 weeks left in the school year.

To set a goal for the student:

- Plot the student's **baseline** score **(6)** on the graph on the first date of the instructional period in which you plan to monitor progress.
- Plot the **end-of-year benchmark (42)** on the graph on the last date of the instructional period in which you plan to monitor progress.
- Draw a goal line from **baseline (6)** to **end-of-year benchmark (42)**.

  Note that the goal line represents the **desired rate of weekly growth**:

  1. Subtract the baseline score from the end-of-year benchmark, to get the amount the student needs to increase to meet the goal. (**42 – 6 = 36**).
  2. Divide this number by the number of weeks in the instructional period (**34 weeks**, so **36/34 = 1.06)**.
  3. Each point on the goal line represents a **1.06 CWS** increase each week.





Is math not your favorite thing?

Hang in there! If you use a district or commercial graphing program, all you need to do is enter the numbers! However, getting the hang of the math can help you better understand your students' data.

OPTION 2: Normative Growth Rates

Identify the average growth rate —or **normative growth rate**— for typically developing students at the grade level at which the student is being monitored. See normative growth rates for Grades 1–3 in the Benchmarking Chart in the CBM Toolkit.

To determine the goal, use the formula:

GOAL = baseline + (desired growth rate × number of weeks)

NORMATIVE GROWTH RATE EXAMPLE

A second-grade student's median score from her first three Picture-Word Prompts is 6 CWS. The district normative growth rate for second-graders is .80 CWS per week. There are 16 weeks left in the instructional period.

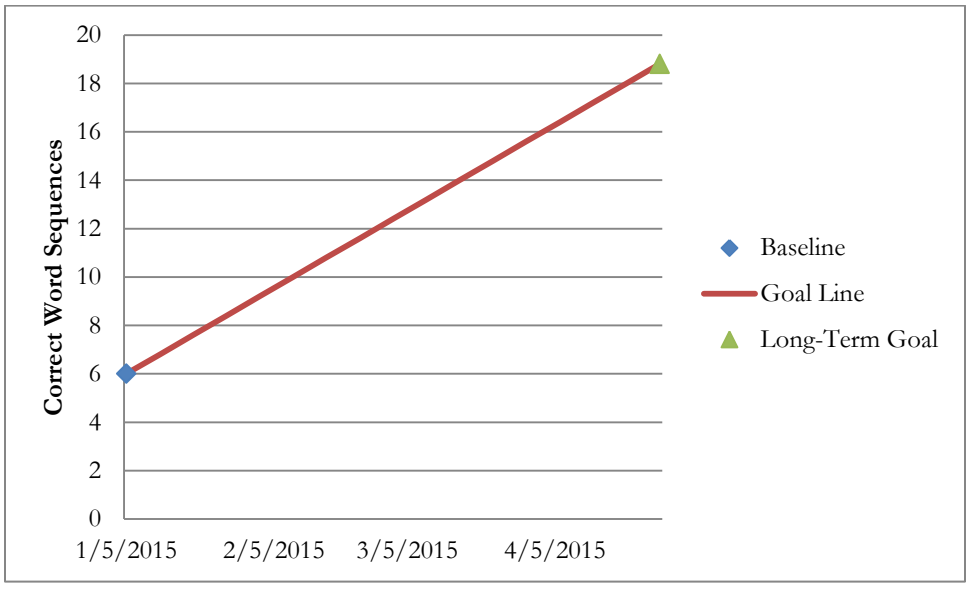
To set a goal for the student:

GOAL = baseline + (desired growth rate × number of weeks)

= 6 + (.80 x 16)
= 6 + 12.8
= 18.8 (round up to 19)

Plot the student's **baseline** score **(6)** on the graph on the first date of the instructional period in which you plan to monitor progress.

- Plot the **long-term goal (19)** on the graph on the last date of the instructional period in which you plan to monitor progress.
- Draw a goal line from **baseline (6)** to **long-term goal (19)**.



ADVANCED OPTION 3: Individualized Student Approach

This option allows you to set a reasonable, ambitious goal for an individual student based on his or her individual performance and growth rate, rather than on an external criterion. This option requires at least eight CBM data points.

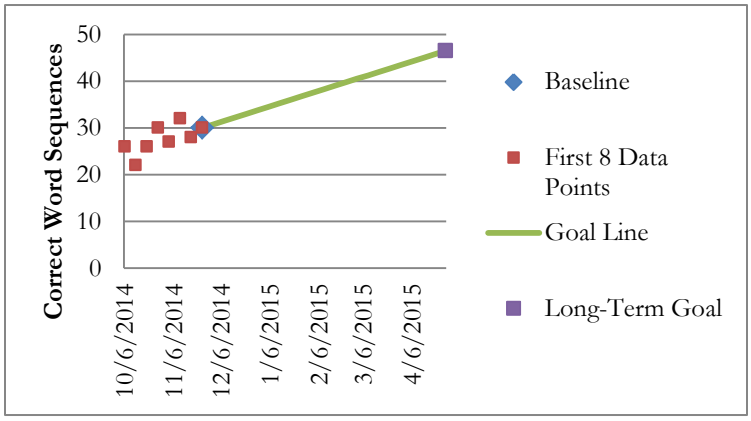
- Identify the **current weekly growth rate** for the target student under baseline conditions, using at least eight CBM data points.
  - Find the median of the first 3 data points (A).
  - Find the median of the last 3 data points (B).
  - Identify the number of weeks in which you've monitored progress so far (W).
  - **Current growth rate (C) = (B – A)/W**.
- Multiply this current weekly growth rate (C) by 1.5. This increases the current growth rate (C) by .5, which leads to an ambitious, but hopefully reasonable, **desired growth rate** (D).
  - **Desired growth rate (D) = C × 1.5**
- To determine the goal, use the formula:
  - GOAL = baseline + (desired growth rate × number of weeks)

INDIVIDUALIZED STUDENT APPROACH EXAMPLE

A student's first 8 CBM scores were 26, 22, 26, 30, 27, 32, 28, and 30 CWS. Baseline is 30 CWS (median of most recent 3 data points); number of instructional weeks remaining is 22.

Current growth rate (C):

- A = Median of first 3 data points (26, 22, 26) = **26**
- B = Median of last 3 data points (32, 28, 30) = **30**
- W = 8 weeks
- C = (B – A)/W
  = (30-26)/8
  = **.5**



Desired growth rate (D):

- D = C × 1.5 = .5 × 1.5 = **.75**

GOAL = baseline + (desired growth rate × number of weeks) = 30 + (.75 × 22) = 30 + 16.5 = **46.5**

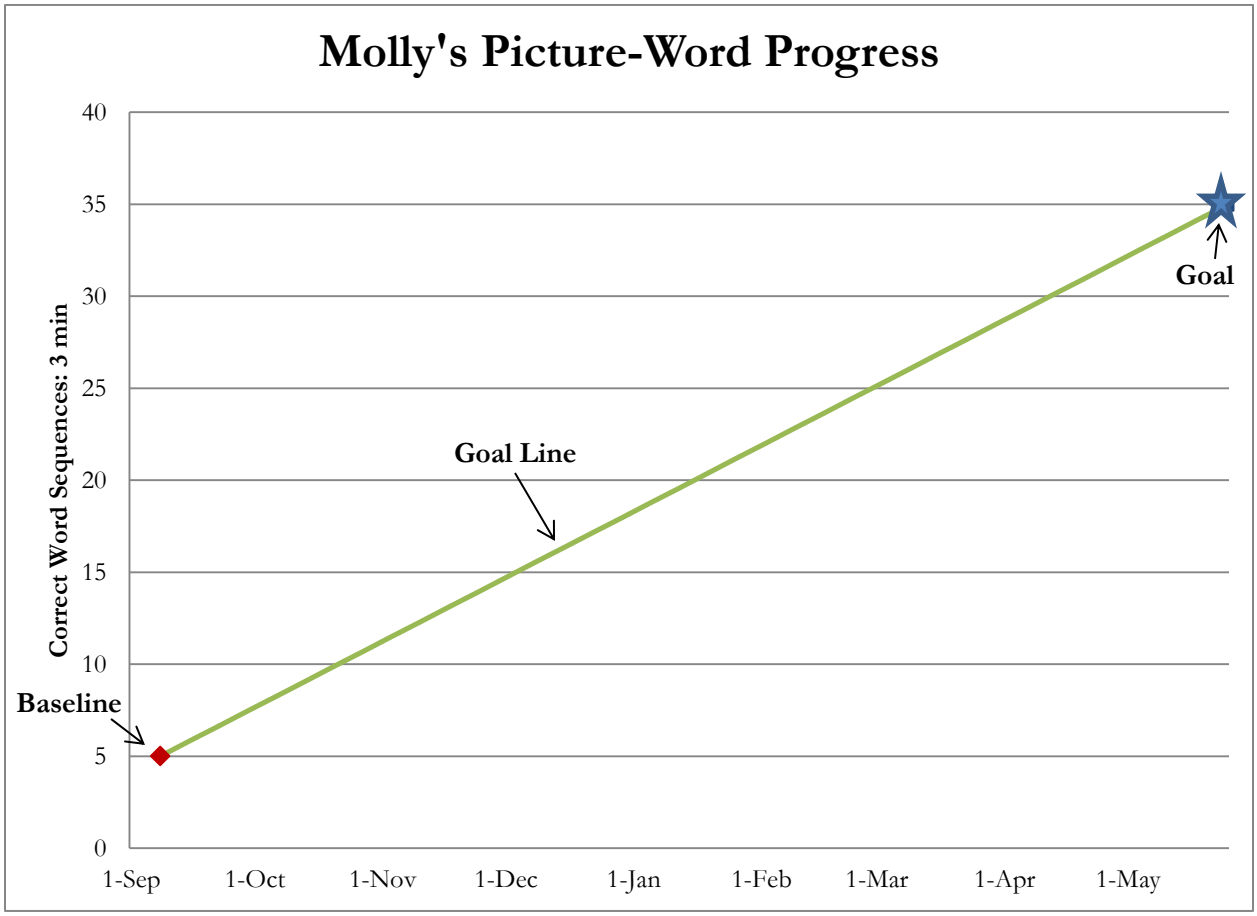
ILLUSTRATION: Mrs. Lewis wants to determine a reasonable, but ambitious, goal for Molly to grow in writing over the school year. She determines a goal based on the school's third grade benchmark for writing in spring.

In this example, the end-of-year (Spring) benchmark for CWS on the Picture-Word Prompt in third grade is 35. Mrs. Lewis determines the **desired rate of growth**:

- She subtracts baseline (**5**) from end-of-year benchmark (**35**).
  35 – 5 = 30.
- She divides **30** by the number of weeks left in the school year (**37**).
  30/37 = .81.

So, each week, Molly will need to increase (on average) **.81 CWS**.

Mrs. Lewis plots the LONG-TERM GOAL on the graph (**35**, indicated by the blue star) at the end of the school year. She draws a GOAL LINE (representing a **.81 weekly increase**) from the BASELINE to the LONG-TERM GOAL. [Click here to view entire case example.]





PRACTICE: Using the baseline established on your Practice Graph in Step 1 (12 CWS), set a goal with the following information:

Baseline = 12
Desired growth rate = .75
Number of weeks = 34

(Practice Graph data and graph are in Appendix C.)

ADDITIONAL PRACTICE: Using the examples below, determine each student's desired growth rate or long-term goal. Then, plot the baseline and goal on a graph (answers are on the next page).

Remember: GOAL = baseline + (desired growth rate × number of weeks)

1. A first grade student's median score from her first three Word Dictation Prompts is 5 CLS. The district end-of-year benchmark for first-graders is 95 CLS. There are 30 weeks left in the instructional period. Use the end-of-year-benchmark to determine the desired growth rate. [click for answer]

2. A second grade student's median score from his first three Picture-Word Prompts is 2 CWS. The district normative growth rate for second-graders is .85 CWS per week. There are 32 weeks left in the instructional period. Use the normative growth rate to determine the goal. [click for answer]

3. ADVANCED OPTION: A third grade student's first eight scores from Story Prompts are as follows: 15, 12, 13, 14, 16, 17, and 18. There are 25 weeks left in the instructional period. Use the individualized student approach to determine the goal. [click for answer]



APPLICATION: Determine a reasonable and ambitious goal for your student, using the option of your choice (benchmark, normative growth rate, or individualized growth rate), and plot this goal on your Student Graph.

ANSWERS TO ADDITIONAL PRACTICE ITEMS:

GOAL = baseline + (desired growth rate × number of weeks)

Answer #1: A first-grade student's median score from her first three Word Dictation prompts is 5 CLS. The district end-of-year benchmark for first-graders is 95 CLS. There are 30 weeks left in the instructional period. Use the end-of-year-benchmark to determine the desired growth rate.

- Subtract the **baseline** score from the **end-of-year benchmark**. This is the number the student needs to gain to reach the benchmark.
    - **95-5 = 90**
- Divide this number by the number of weeks remaining in the instructional period to get a **desired growth rate**.
    - **90/30 = 3 CLS per week**

Answer #2: A second-grade student's median score from his first three Picture-Word prompts is 2 CWS. The district normative growth rate for second-graders is .85 CWS per week. There are 32 weeks left in the instructional period. Use the normative growth rate to determine the goal.

Baseline = 2
Desired rate = .85
GOAL = baseline + (desired growth rate × number of weeks)
= 2 + (.85 × 32)
= 2 + 27.2
= 29.2

Answer #3: A third-grade student's first eight scores from Story prompts are as follows: 15, 12, 11, 14, 16, 15, and 14. There are 25 weeks left in the instructional period. Use the individualized student approach to determine the goal.

Current growth rate (C):
A = Median of first 3 data points (15, 12, 11) = 12
B = Median of last 3 data points (16, 15, 14) = 15
C = B – A = 15 - 12 = 3
Desired growth rate (D):
D = C × 1.5 = 3 × .5 = 1.5
GOAL = baseline + (desired growth rate × number of weeks)
= 12 + (1.5 × 25) = 12 + 37.5 = 49.5

Step 3: Implement high quality instruction with fidelity based on student needs.



Toolkit needed: Intervention



DESCRIPTION: To implement DBI effectively, it is critical that you implement high quality instruction with fidelity—by following the instructional protocol according to methods demonstrated to be effective. If your instruction is not implemented with fidelity, it will be difficult to determine whether a student is not responding because the instruction is not effective, or because the instruction was not implemented properly. Below is a description of how to go about selecting, planning, and implementing a high quality instructional program in beginning writing.

After you have set the student's goal, we recommend that you use the **Writing Intervention Toolkit** to do the following:

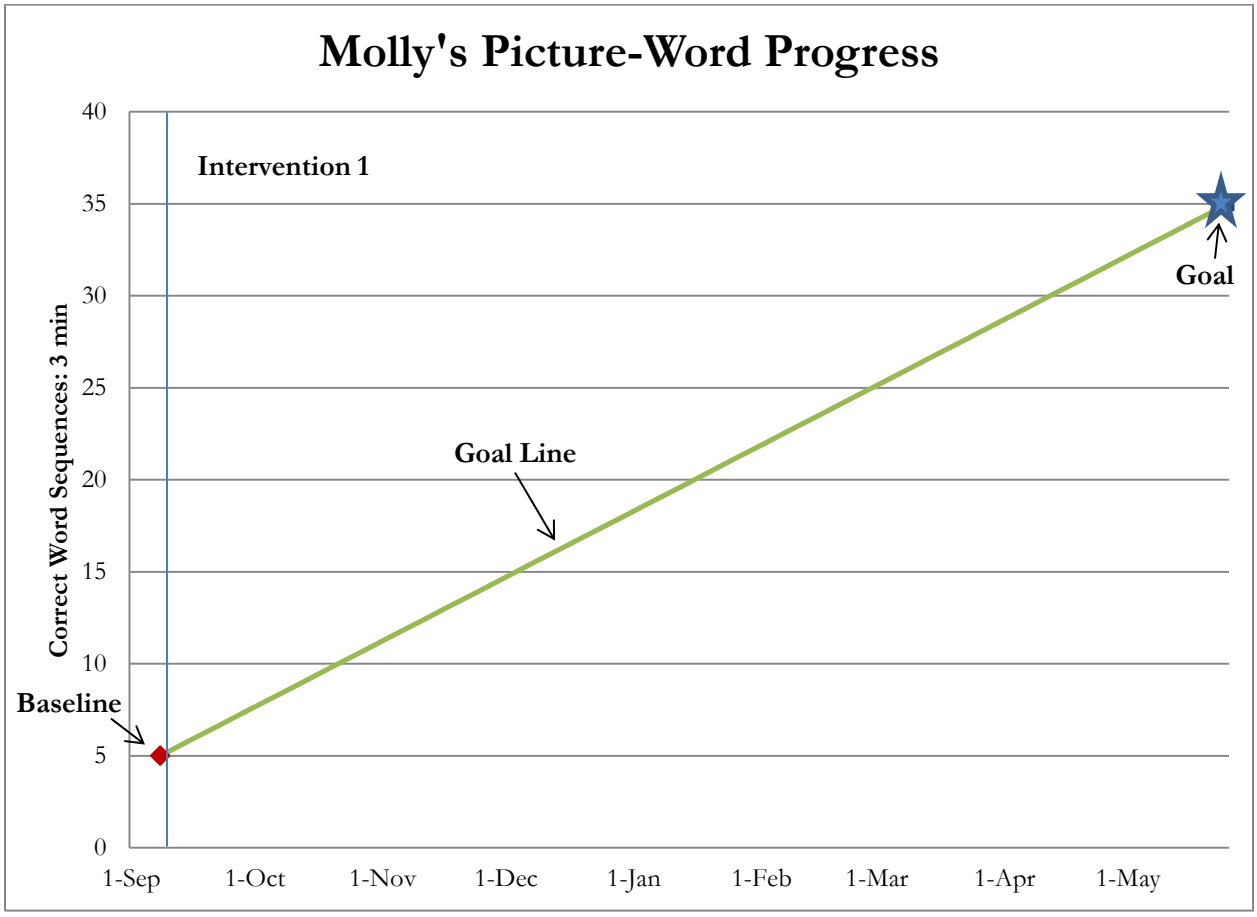
a. Select research-based instructional methods to implement based on student strengths and needs.
   - For guidance on determining the *focus* and *content* of instruction, see *Instructional Alignment Tool* and *Diagnostic Checklists* in the Writing Intervention Toolkit.
   - For a comprehensive list of research-based writing intervention studies, see *Writing Interventions References* in the Writing Intervention Toolkit.

b. Determine the *dosage* (duration and frequency) of writing instruction. For example, you may decide to implement the intervention *3 times per week for 30 min.* When possible, use guidelines from research to determine the duration and frequency.

c. Develop a Writing Instructional Plan (WIP) based on your selected instructional approaches. The Writing Intervention Toolkit includes:
   - Basic WIP templates (for Transcription, Transcription + Text Generation, and Self-regulation)
   - Filled-In WIP samples(for Transcription, Transcription + Text Generation, and Self-regulation)
   - Skill-based mini-lessons to be used within the WIP or separately, as needed

d. Implement the core components of the intervention with *fidelity,* by following the intervention protocol according to methods demonstrated to be effective.

ILLUSTRATION: Mrs. Lewis draws a vertical line after the baseline, and labels the new section INTERVENTION 1. She consults the *Instructional Alignment Tool* and the *Diagnostic Checklists* (in Writing Intervention Toolkit) and decides to focus on handwriting, spelling, and sentence construction—skills determined by Molly's IEP team to be critical for Molly's success in school. She uses these skills to design a *Writing Instructional Plan* (see Writing Intervention Toolkit). [Click here to view entire case example.]



PRACTICE: On your Practice Graph, insert a vertical line after the baseline data to indicate the beginning of Intervention 1. (Practice Graph data and graph are in Appendix C.)

APPLICATION: On your Student Graph, insert a vertical line after the baseline data to indicate the beginning of Intervention 1.

Step 4: Monitor student progress toward the goal.



Toolkit needed: CBM



DESCRIPTION: While implementing high quality instruction, it is critical to monitor the student's progress continually toward the goal to determine whether progress is adequate and whether the instructional program is effective. To monitor progress, you will need to:

a. Determine a schedule for progress monitoring, including the *frequency* of progress monitoring, the *number of forms*, and the *type of score to graph*. Select from the following options:

Frequency	Number of Forms	Type of Score to Graph
1 x per week (recommended)	***1 form per session.***	***Single score from each form.***
Every other week	2-3 forms per session	Mean or median of 2-3 scores.

b. Graph the data regularly. Your use of CBM for DBI will be most effective and efficient if you regularly graph and examine your student's progress.

c. When at least 8 CBM scores have been collected and graphed, add a *trend line* that represents the student's actual rate of progress. (At least 8 scores are needed to obtain a reliable trend line; e.g., McMaster et al., 2011).

The trend line can be added in the following ways:

- **Using a spreadsheet program such as Microsoft Excel**: You can use the program to add a trend line. (See tutorial in CBM Toolkit.)
- **Using a commercial or district progress monitoring program**: The trend line may be plotted automatically.

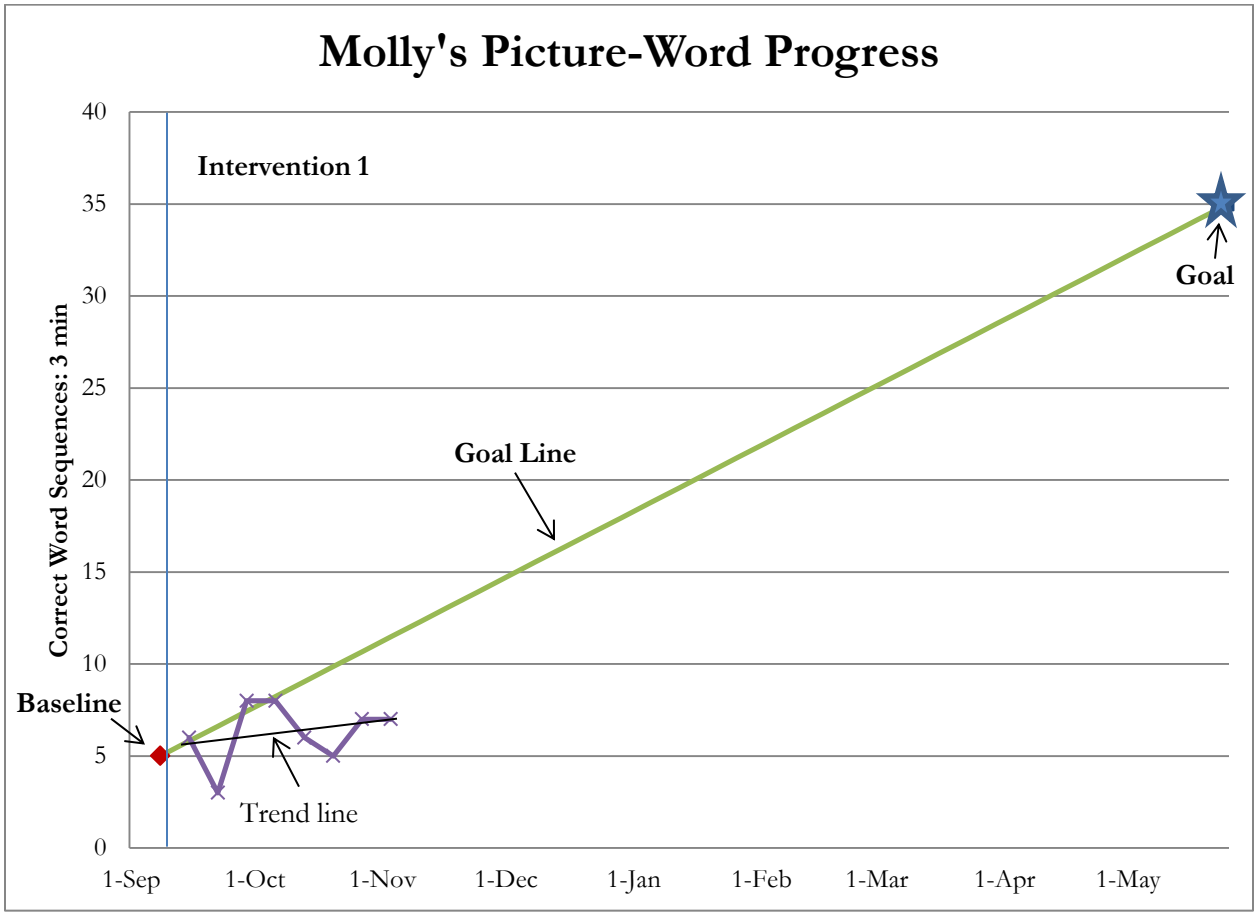


How do I manage all these data??

It's worth taking the time to develop a system for administering CBM and scoring and entering your data on a regular basis. The more you make CBM part of your routine, the easier and more efficient it will become. We suggest the following:

- Gather all materials you'll need (administration and scoring procedures, student folders with copies of prompts, timer, pencils) and keep them in one place so that they are ready to go when it's time to do CBM.
- Keep a folder for each student, with copies of blank writing prompts in the left pocket.
- Score prompts as soon as possible, enter the data immediately, and file the completed prompts in the right pocket of the student's folder.

ILLUSTRATION: Mrs. Lewis monitors Molly's progress toward the goal by giving a 3-minute CBM Picture-Word prompt each week. She continues to score each prompt using CWS. She plots Molly's progress monitoring score each week in the INTERVENTION 1 section on the graph. After collecting and plotting multiple data points (best practice suggests 8 data points or weekly progress monitoring scores), Mrs. Lewis adds a trend line. [Click here to view entire case example.]





PRACTICE: On your Practice Graph, plot the following 8 data points. Then, insert a trend line to show the student's rate of progress. (Practice Graph data and graph are in Appendix C.)

Date	Baseline	Intervention 1
9/1/2014	12	
9/8/2014		12
9/15/2014		10
9/22/2014		15
9/29/2014		11
10/6/2014		14
10/13/2014		12
10/20/2014		13
10/27/2014		14



APPLICATION: Collect weekly CBM-W data for your student, for at least 8 weeks. Plot the data on your Student Graph. Then, insert a trend line to show the student's rate of progress.

Step 5: Use decision rules to evaluate student progress & instructional effectiveness.

Toolkit needed: DECISION-MAKING

DESCRIPTION: CBM graphs are used to evaluate student progress and to formulate instructional decisions. Standard CBM decision rules guide decisions about the adequacy of student progress and the need to revise goals and instructional programs.

To make decisions based on CBM data, first consider the following three features of the student's performance and progress:

- **Level:** Examine the student's level in relation to the goal line. Are most of the student's data points on or above the goal line (as in Panel A of Figure 8)? Or are most of them below the goal line (as in Panel B of Figure 8)?

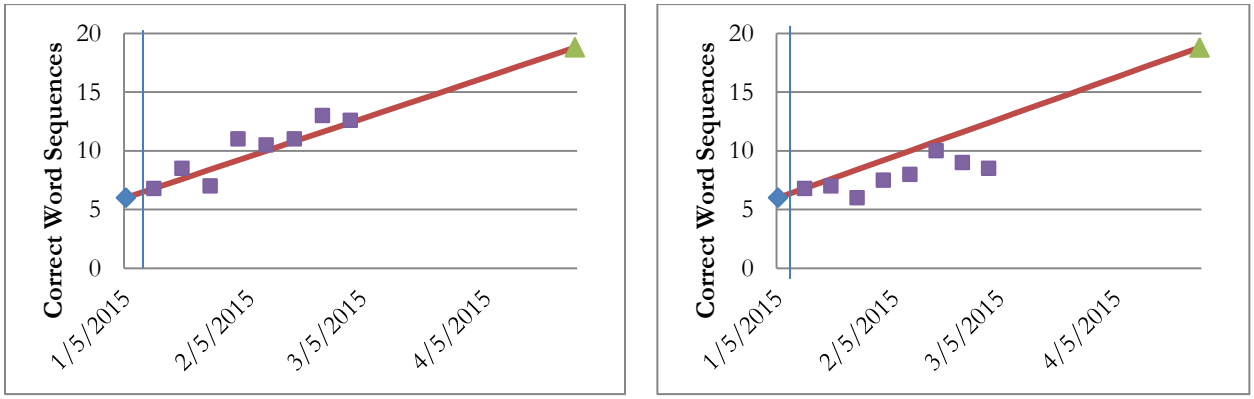


Figure 8. ***Most data points on or above (Panel A) or below (Panel B) the goal line.***

- **Trend:** Does the student's rate of growth (shown by the *trend line*, as described in Step 4) appear to be sufficient to meet the long-term goal (see Figure 9 below)?

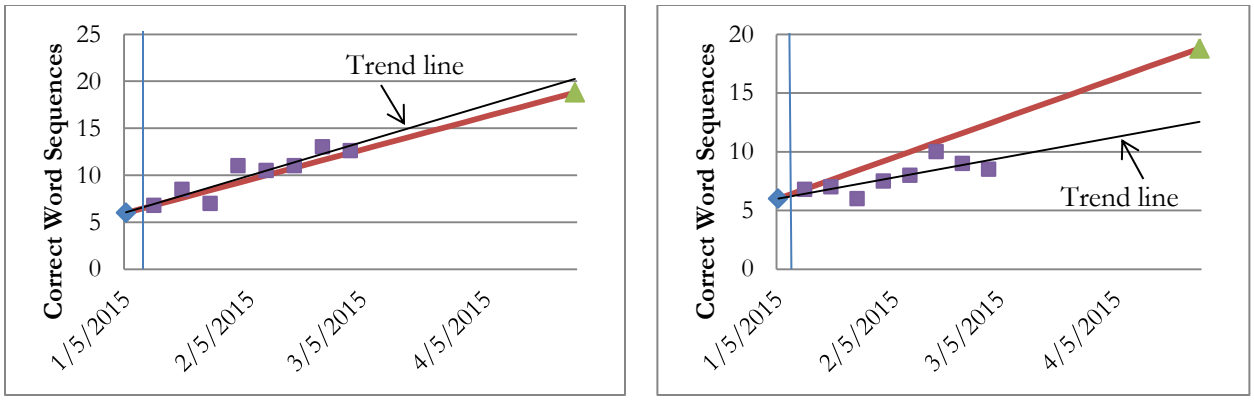


Figure 9. ***Trend line added to data.***

- **Variability:** Is there a lot of variability in the scores (for example, extreme highs and/or lows, as in Figure 10)? If so, be wary of the effect extreme scores might have on the trend line. Panels A and B in Figure 10 are the same as Panels A and B in Figure 9, but with one score altered. Note the impact these extreme low and extreme high scores have on the trend line. Consider dropping extreme scores, or continuing to collect data until you have a more stable trend line.

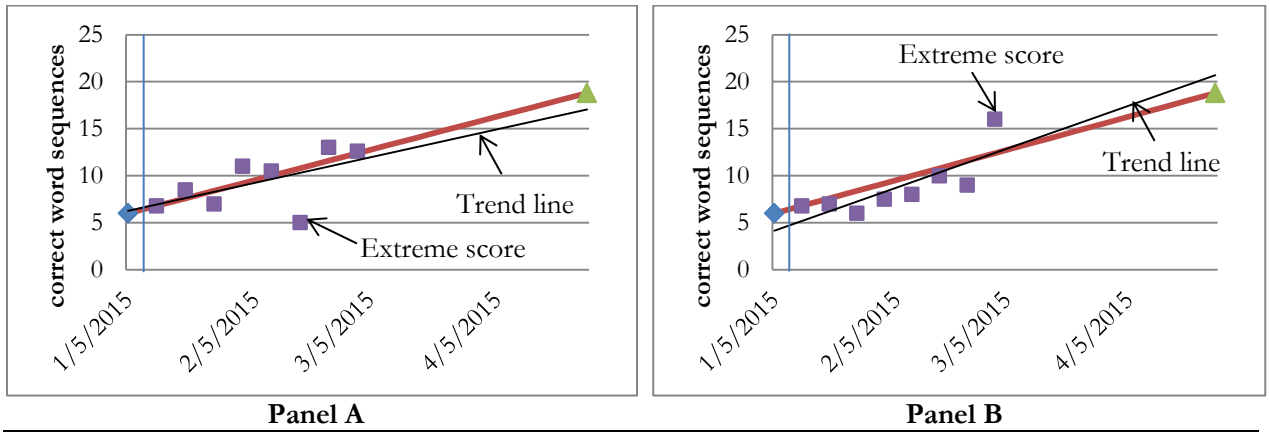


Figure 10. ***One extremely low score lowers the trend line in Panel A, and one extremely high score increases the trend line in Panel B.***

If the data are stable enough (not too variable) to make a decision, examine the student's trend line to make one of three decisions:

DECISION 1: KEEP AS-IS:

This decision is made when the student's trend line is in line with the goal line, suggesting that the instructional program is effecting satisfactory student progress (see Figure 13).

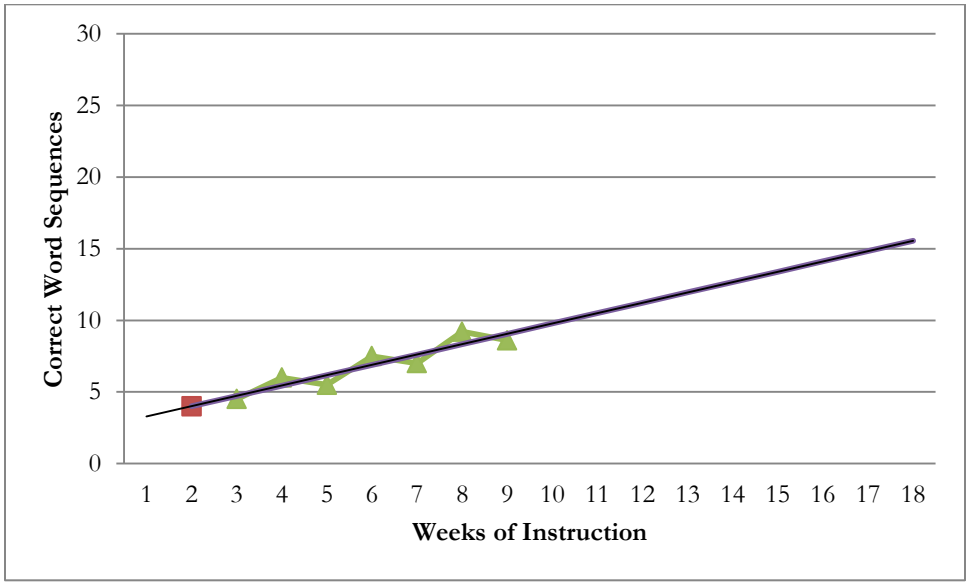


Figure 13. ***Trend Line Matches Goal Line***

If the trend line matches the goal line, the decision is to KEEP AS-IS. Re-evaluate the student graph after four additional data points to determine whether to raise the goal, change instruction, or keep as-is.

DECISION 2: INCREASE THE GOAL:

This decision is made when the student's trend line is *steeper* than expected in relation to the goal line (see Figure 11). The goal is increased to maintain high expectations of the student and to maintain a rigorous instructional program.

The new goal can simply be an extension of the student's current trend line (see Figure 11).

OR, you can use the formula:

GOAL = *median of last 3 data points* + (*desired growth rate* × *number of weeks*)

- *Desired growth rate* can be the same growth rate used before (or, if you wish to increase the growth rate, multiply the original growth rate by .5).
- *Number of weeks* is number of weeks remaining in the instructional period.

Plot the new goal and goal line on the graph.

If the student's current goal is already the end of year benchmark for his/her grade, then continue with the current goal but consider lessening intensity of intervention (e.g., by increasing group size, reducing frequency/duration).

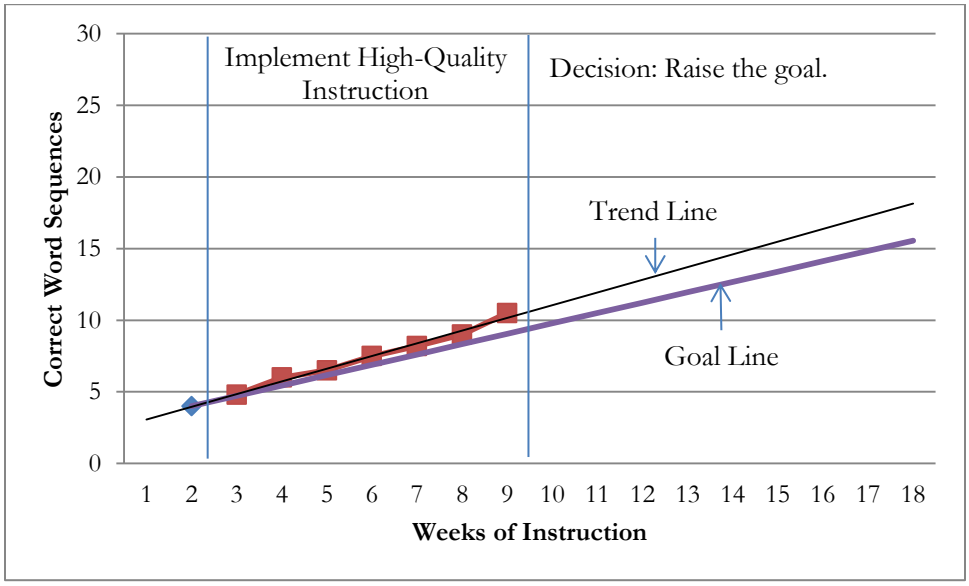


Figure 11. ***Trend Line Above Goal Line.***

On this graph, the trend line is steeper than the goal line. Therefore, you should RAISE THE GOAL to boost the actual rate of student progress. Insert a vertical line to indicate the point where you raised the student's goal. You should re-evaluate the student graph after collecting 4 more data points to determine whether the student's new goal is appropriate or whether an instructional change is needed.

DECISION 3: CHANGE INSTRUCTION:

This decision is made when the student's trend line is *less steep* than expected in relation to the goal line (see Figure 12). Instruction is changed because the existing instructional program is not sufficient to increase student progress.

Note that the decision should rarely, if ever, be to lower the goal.

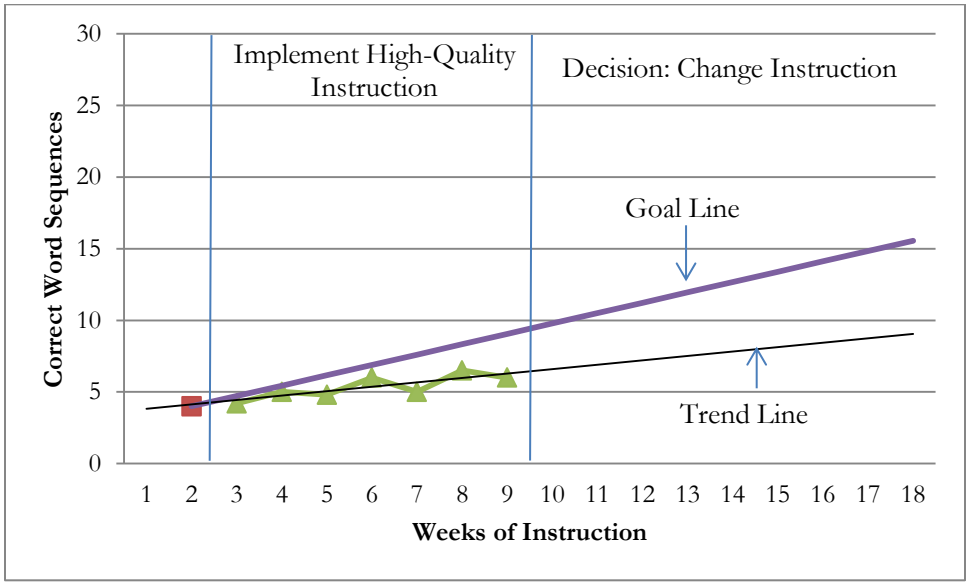
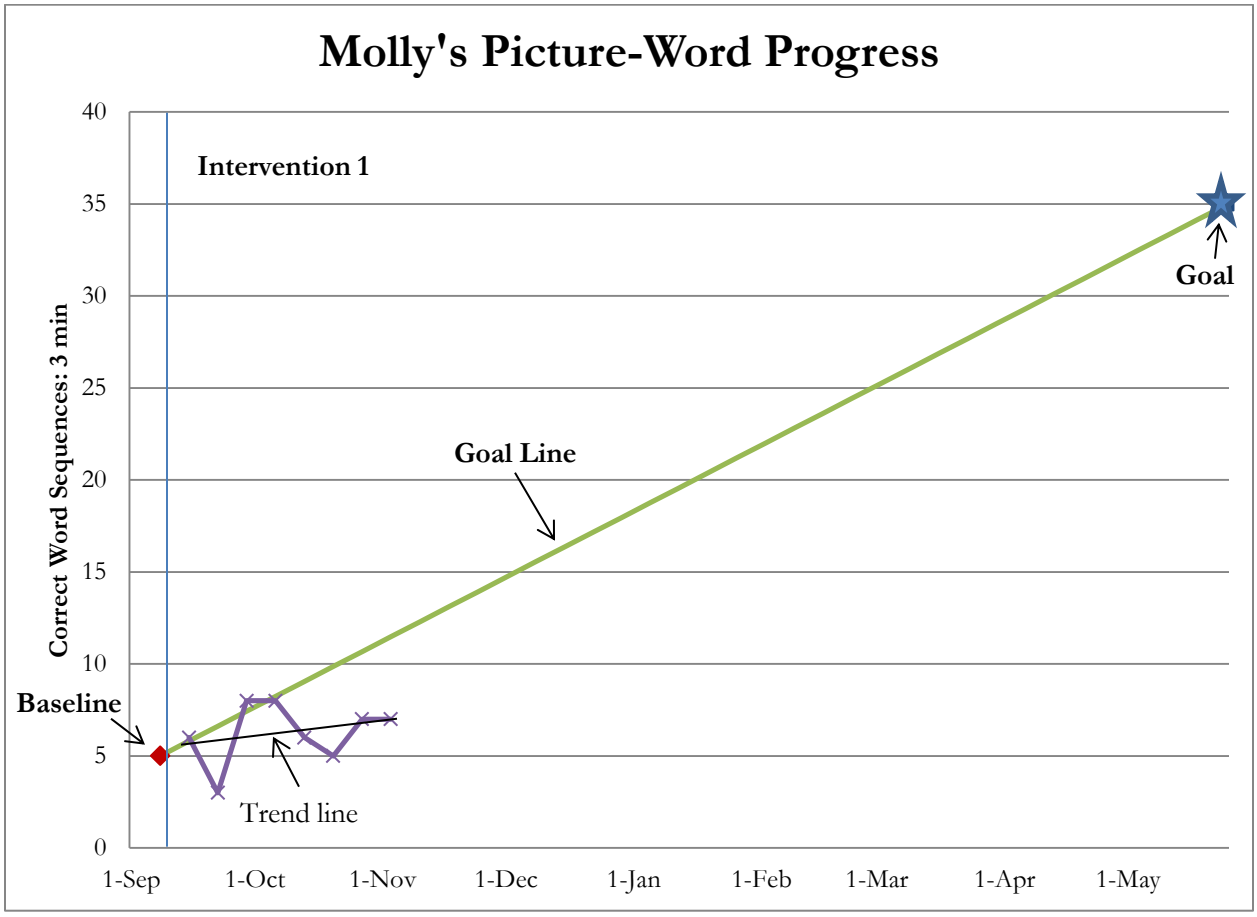


Figure 12: ***Trend Line below Goal Line***

On this graph, the trend line is flatter than the goal line. Therefore, you should CHANGE INSTRUCTION. Note that the initial decision should NOT be to lower the goal or the goal line. The instructional program should be tailored to bring a student's scores up so they match or surpass the goal line. Draw a vertical line to indicate the point where you made an instructional change. Re-evaluate the student graph after 4 data points to determine whether the change was effective.

ILLUSTRATION: Mrs. Lewis uses a decision rule that tells her to compare Molly's current rate of growth (trend line) to the desired rate of growth (goal line). When Mrs. Lewis compares Molly's trend line to her goal line, she sees that Molly's progress falls below the goal line. She decides an instructional change is needed. [Click here to view entire case example.]



PRACTICE: Look at the data you plotted on your Practice Graph for Intervention 1. Make one of the following decisions: (1) INCREASE THE GOAL, (2) CHANGE INSTRUCTION, or (3) KEEP AS IS. (Practice Graph data and graph are in Appendix C.)

APPLICATION: Look at the data you plotted on your Student Graph for Intervention 1. Make one of the following decisions: (1) INCREASE THE GOAL, (2) CHANGE INSTRUCTION, or (3) KEEP AS IS.

Step 6: Generate hypotheses to individualize instruction.



Toolkit needed: DECISION-MAKING



DESCRIPTION: When you have determined that an instructional change is needed, you will need to generate hypotheses about why the student is not making sufficient progress and what instructional changes to make to address the problem.

Use the **Decision-Making Rubric** in the Decision-Making Toolkit as a guide for making systematic decisions about what instructional changes to consider. This rubric prompts you to think about the questions in Table 3 below, which may lead to specific hypotheses.

Table 3
Data-Based Decision-Making: Questions, Possible Hypotheses, and Instructional Considerations

Question	Possible Hypotheses	Instructional Considerations
How was the intervention chosen and was it delivered as intended? • Is the intervention supported in research for students with similar needs as my students? • Have I implemented the intervention with fidelity?	The student's progress will increase if I implement a research-based intervention with fidelity.	• Make sure I am implementing an appropriate research-based intervention. • Adjust my fidelity of implementation.
Does the student need changes to setting, format, or delivery of the intervention?	• The student needs more time. • The student is not motivated. • The student needs more individualized attention. • The student needs more opportunities to practice. • The student needs more explicit instruction.	• Increase the frequency and/or duration of intervention. • Add a motivational component. • Decrease the group size. • Increase opportunities to respond. • Increase modeling and guided practice with corrective feedback.
Does the student need changes to the CONTENT of the intervention?	• The student needs to work on (specific letters, words, vocabulary, text structure, etc.)	• Use the *Transcription Scope and Sequence* tool in the Writing Intervention Toolkit to select appropriate intervention content, or other scope and sequence materials from your curriculum.
Does the student need changes to the FOCUS of the intervention?	• The student needs more emphasis on (Transcription? Text Generation? Self-regulation?)	• Use the *Diagnostic Checklists* in the Writing Intervention Toolkit to select an appropriate focus.



ILLUSTRATION: Mrs. Lewis generates several hypotheses about why Molly is making insufficient progress. Her hypotheses are based on her examination of Molly's progress monitoring data collected during intervention, and her use of the *Decision-Making Rubric* (in Decision-Making Toolkit). After considering several hypotheses, she chooses the one that suggests that Molly needs more time in her current intervention. [Click here to view entire case example.]



PRACTICE: Consider the following scenarios, in which each student's data indicate insufficient progress to meet the long-term goal. Develop a hypothesis about each student's needs.

Scenario *(Click on the student's name to view a possible hypothesis. There is likely more than one possible hypothesis.)*
1. Ana's teacher has been implementing a research-based intervention with fidelity, with a group of 5 students with similar needs in terms of *focus* (text generation) and *content* (writing descriptive paragraphs). Ana is quiet and tends to get lost in the chaos of the four other more boisterous group members.
2. Demitri's teacher has been implementing a research-based intervention with fidelity. Demitri's teacher carefully matched the intervention to his needs in terms of *focus* (spelling and text generation) and *content* (multi-syllabic words). Demitri seems motivated to do well but just doesn't seem to be getting what his teacher is asking him to do.
3. Emily's teacher has been implementing a research-based intervention that she learned about at a workshop. Instead of implementing it according to the manual, the teacher has put her own spin on it—she has added, changed, and eliminated parts to "make it her own."
4. Evan's teacher has been implementing a research-based intervention with fidelity. Evan's teacher carefully matched the intervention to his needs in terms of *focus* (self-regulation) and *content* (writing personal narratives). Evan has demonstrated occasionally that he *can* perform well, but he frequently complains that he hates writing and that it is boring.
5. Horatio's teacher has been implementing an intervention she found on the internet. No published research supports the efficacy of the intervention.
6. Jackson's teacher has been implementing a research-based intervention with fidelity in a small group of 5 students who struggle with spelling. She is currently focusing on CVC and CVCe words. The intervention includes a motivational component (goal setting with incentive) that typically works well with Jackson. However, Jackson frequently complains that the words are "too easy" and "boring." His teacher notices that he typically does spell the target words more quickly and accurately than the other group members do, but she doesn't feel that she can change his group because she only has 30 minutes with him each day, and this is the only time that works in her schedule.
7. Ming's teacher has been implementing a research-based intervention with fidelity, focusing on the writing process (planning, organizing, writing, and revising). Ming has lots of good ideas to write about, but she has lots of difficulty getting her ideas on paper. She spends most of her time trying to figure out how to spell the words, and painstakingly tries to form the letters. She often gets frustrated and scribbles out her work, and often ends up wadding up her paper and throwing it away.
8. Tameka's teacher has been implementing a research-based intervention with fidelity. Tameka's teacher carefully matched the intervention to her needs in terms of *focus* (transcription) and *content* (forming all letters of the alphabet; spelling CVC words). The teacher spends 15 minutes, 3 times per week, working one-on-one with Tameka. Tameka is very motivated to do well and has been making some progress, but not sufficient to meet her goal.

Possible Hypotheses *(Click on the student's name to return to the scenario.)*

Ana may benefit from more individualized attention during intervention.

Dimitri may benefit from more explicit instruction.

Emily may benefit from the intervention if it is implemented with fidelity.

Evan may benefit from a motivational component.

Horatio may benefit more from a research-based intervention.

Jackson may benefit from working on different word patterns that are more closely matched to his needs.

Ming may benefit from an intervention focused on transcription (handwriting and spelling) rather than the writing process.

Tameka may benefit from more time in intervention OR Tameka may benefit from more opportunities to practice.



APPLICATION: Based on the available data and other information you have about your student, develop a hypothesis about his or her instructional needs.

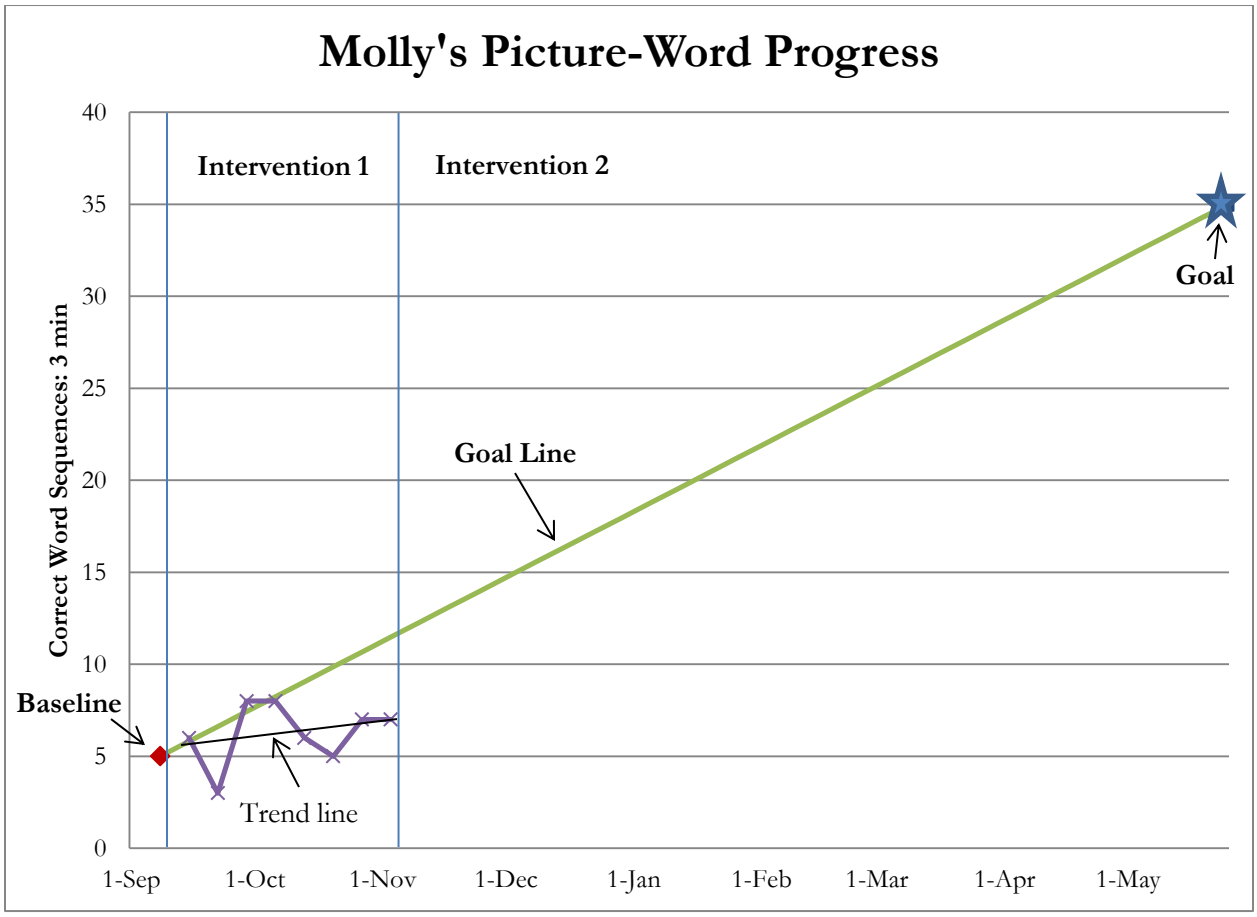
Step 7: Make instructional changes based on hypotheses.

Toolkits needed: Decision-Making and Intervention

DESCRIPTION: Using the hypothesis you developed about the student's needs, identify the instructional change(s) that you will make (using the Decision-Making Rubric to identify possible changes), and document these changes on the *Decision Log* form found in the Decision-Making Toolkit.

Documenting the changes will allow you to track your changes systematically over time, to evaluate whether your hypothesis was correct or whether different changes are needed. This documentation will also provide a useful record for special education referral and/or Individualized Educational Plan (IEP) meetings.

ILLUSTRATION: Mrs. Lewis increases the frequency of intervention from 3 to 5 days per week. She draws another vertical line after INTERVENTION 1 and labels the new section INTERVENTION 2. She records the change on her *Decision Log* (in Decision-Making Toolkit).
[Click here to view entire case example.]





PRACTICE: Using the scenarios in Step 6, the hypotheses you generated, and the Decision-Making Rubric, consider a possible change that the teacher could consider making to the intervention.

Scenario	**Hypothesis (Step 6 Practice)**	**Possible Change (Step 7 Practice)**
Ana's teacher has been implementing a research-based intervention with fidelity, with a group of 5 students with similar needs in terms of *focus* (text generation) and *content* (writing descriptive paragraphs). Ana is quiet and tends to get lost in the chaos of the four other more boisterous group members.	Ana may benefit from more individualized attention during intervention.	Provide more individualized attention, OR a smaller group size, OR a different group (if possible).
Demitri's teacher has been implementing a research-based intervention with fidelity. Demitri's teacher carefully matched the intervention to his needs in terms of *focus* (spelling and text generation) and *content* (multi-syllabic words). Demitri seems motivated to do well but just doesn't seem to be getting what his teacher is asking him to do.	Dimitri may benefit from more explicit instruction.	Provide more modeling, guided practice, and immediate corrective feedback.
Emily's teacher has been implementing a research-based intervention that she learned about at a workshop. Instead of implementing it according to the manual, the teacher has put her own spin on it—she has added, changed, and eliminated parts to "make it her own."	Emily may benefit from the intervention if it is implemented with fidelity.	Implement the intervention with fidelity.
Evan's teacher has been implementing a research-based intervention with fidelity. Evan's teacher carefully matched the intervention to his needs in terms of *focus* (self-regulation) and *content* (writing personal narratives). Evan has demonstrated occasionally that he *can* perform well, but he frequently complains that he hates writing and that it is boring.	Evan may benefit from a motivational component.	Add a goal setting component with incentive.
Horatio's teacher has been implementing an intervention she found on the internet. No published research supports the efficacy of the intervention.	Horatio may benefit more from a research-based intervention.	Find and implement a research-based intervention that fits Horatio's specific needs.

Scenario	**Hypothesis (Step 6 Practice)**	**Possible Change (Step 7 Practice)**
Jackson's teacher has been implementing a research-based intervention with fidelity in a small group of 5 students who struggle with spelling. She is currently focusing on CVC and CVCe words. The intervention includes a motivational component (goal setting with incentive) that typically works well with Jackson. However, Jackson frequently complains that the words are "too easy" and "boring." His teacher notices that he typically does spell the target words more quickly and accurately than the other group members do, but she doesn't feel that she can change his group because she only has 30 minutes with him each day, and this is the only time that works in her schedule.	Jackson may benefit from working on different word patterns that are more closely matched to his needs.	Conduct a spelling inventory and select content that is better matched to Jackson's current skills.
Ming's teacher has been implementing a research-based intervention with fidelity, focusing on the writing process (planning, organizing, writing, and revising). Ming has lots of good ideas to write about, but she has lots of difficulty getting her ideas on paper. She spends most of her time trying to figure out how to spell the words, and painstakingly tries to form the letters. She often gets frustrated and scribbles out her work, and often ends up wadding up her paper and throwing it away.	Ming may benefit from an intervention focused on transcription (handwriting and spelling) rather than the writing process.	Add handwriting and spelling component to Ming's instructional plan.
Tameka's teacher has been implementing a research-based intervention with fidelity. Tameka's teacher carefully matched the intervention to her needs in terms of *focus* (transcription) and *content* (forming all letters of the alphabet; spelling CVC words). The teacher spends 15 minutes, 3 times per week, working one-on-one with Tameka. Tameka is very motivated to do well and has been making some progress, but not sufficient to meet her goal.	Tameka may benefit from more time in intervention. OR Tameka may benefit from more opportunities to practice.	Increase Tameka's intervention time to 30 minutes (or to 4-5 times per week). OR Within the 15 minute intervention, decrease the amount of teacher talk and increase Tameka's opportunities to practice.



APPLICATION: Based on the hypothesis you generated in Step 6, consider a possible change you could make to your student's intervention. Discuss your ideas with a colleague.

Step 8: Repeat steps 4 through 7.



Toolkit needed: Decision-Making



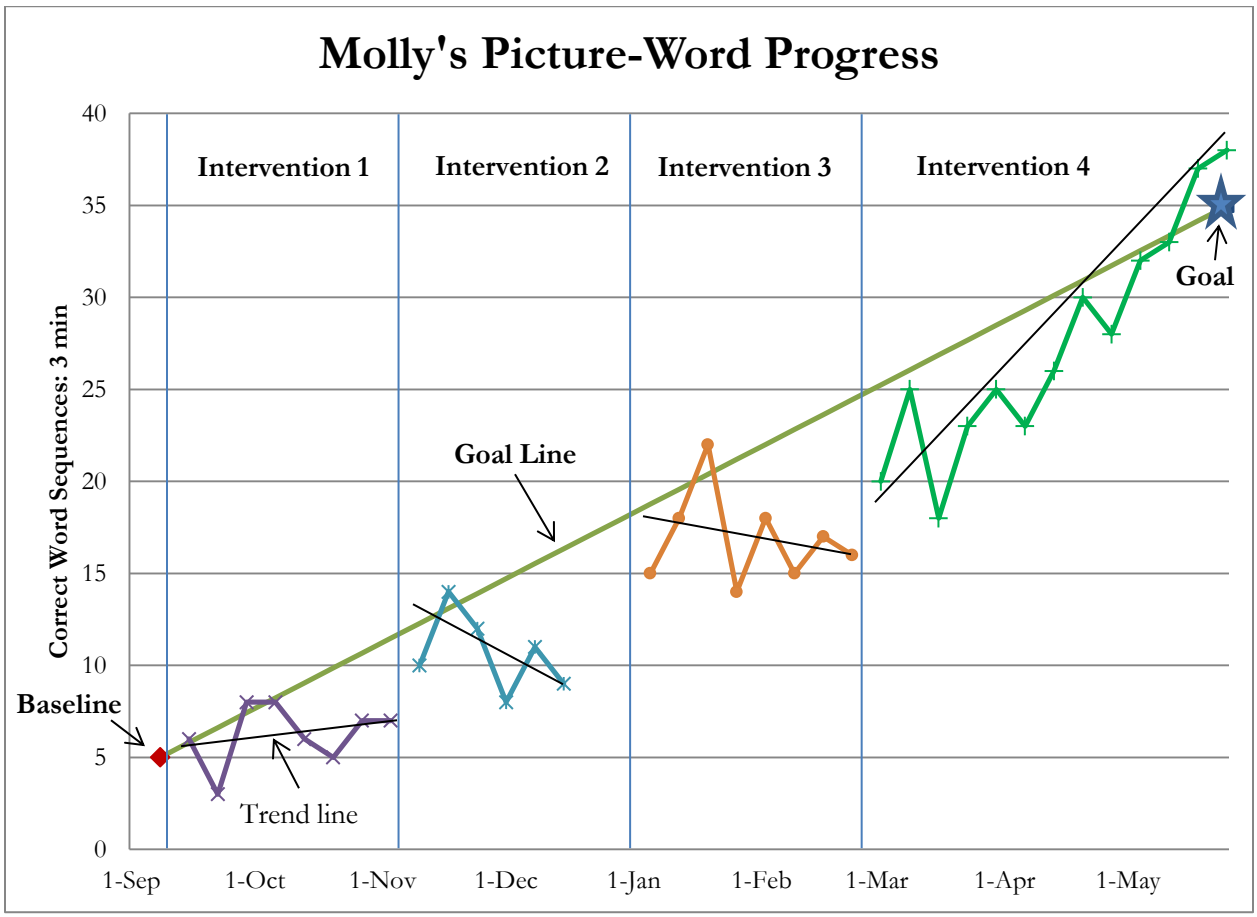
DESCRIPTION: DBI will only be successful if you continue to implement the process with fidelity—as described in this manual:

- Repeat Step 4 by continuing to monitor the student's progress.
- Repeat Step 5 by using decision rules to evaluate student progress and instructional effectiveness.
- Repeat Step 6 by generating hypotheses about the types of instructional changes that may be needed, using the *Decision-Making Rubric* as a guide.
- Repeat Step 7 to make instructional changes based on these hypotheses. Record your changes on the Decision Log form so that you are systematically tracking each change.
- You can use the *Fidelity Tools* (provided in each toolkit) to monitor your fidelity to the DBI process.



ILLUSTRATION: After collecting several weekly data points in Intervention 2, it is clear that increasing the frequency of instruction is not sufficient to improve Molly's progress. Mrs. Lewis continues to use the *Decision-Making Rubric* (in Decision-Making Toolkit) and generates a new hypothesis—that Molly needs more explicit instruction with additional modeling and practice. She adds this change to her instruction, and continues to monitor Molly's progress during INTERVENTION 3.

During Intervention 3, Molly's progress increases; however, her performance is still below the goal line. Thus, Mrs. Lewis develops a new hypothesis—that Molly needs additional reinforcement for working on her writing, and would benefit from monitoring and charting her own progress toward her goal. She adds this change to her instruction, and continues to monitor Molly's progress during INTERVENTION 4. Finally, Molly meets and exceeds her goal. Now Mrs. Lewis can re-visit the *Instructional Alignment Tool* and *Diagnostic Checklists* (in Writing Intervention Toolkit) to establish her next area of focus for Molly. [Click here to view entire case example.]





PRACTICE: Examine the data and trend lines in each intervention phase of Molly's graph. Do you agree with each decision the teacher made? Why or why not?

Now add the following data to your practice graph. What is your decision? Why? Practice Graph data and graph are in Appendix C.)

Date	Intervention 2
11/3/2014	18
11/10/2014	20
11/17/2014	25
11/24/2014	22
12/1/2014	24
12/8/2014	23
12/15/2014	27



APPLICATION: Continue to monitor your student's progress, making systematic adjustments to your instruction over time, based on data and your hypotheses about your student's needs.

Communicating With Parents/Guardians

CBM data can be used to communicate clearly with parents. CBM is particularly useful for communication with parents because the graphed data provide a 'picture' of student progress, which can be used to clearly communicate student growth or lack thereof.

Parents need to clearly understand what all of the pieces of the graph mean (baseline, goal and goal line, data points and trend, and so on), so it is best to meet with parents face to face the first time you introduce their child's graphed data. Providing a labeled sample graph can aid in communicating basic features. Following this introduction, let parents know how often they can expect to receive graphs and how they will receive them. For example, you could email or send a hard copy of the graph home weekly, or you could provide the graph once per month instead. Making sure that students understand their results is important as well so that they can communicate with their parents about their progress.

What are some ways that schools have communicated with parents about the DBI process? In some cases, the school will have a session on the basic components of the model at back to school night, open house, or at parent-teacher conferences. Some schools choose to send a standard letter home to all parents of students who are participating in the DBI process. Teachers may help their students conduct student-led conferences for their parents, using CBM data to support the process.

However you choose to share information, please include communication with parents as a key piece of your DBI process. This step provides yet another opportunity for the process to be implemented and used with the greatest potential effects.

Supporting Student Progress within Multi-Tiered Systems of Support (MTSS)

Currently, many schools and districts are adopting multi-tiered systems of support (MTSS) to address the academic and behavioral needs of all students. Within such systems, Response to Intervention (RTI) is an increasingly common approach used to address students' academic needs, particularly in reading and mathematics. RTI can also be applied to students' needs in writing. We believe that the CBM procedures presented in this manual are well-suited to conducting RTI with students who are at risk for experiencing writing difficulties in school.

RTI involves multiple tiers of assessment and intervention that include

(a) universal screening to identify students at risk,
(b) evidence-based core instruction implemented with fidelity,
(c) ongoing progress monitoring, and
(d) increasingly intensive, individualized intervention for students for whom core instruction is not sufficient.

Many of the key features needed to implement RTI in early writing are detailed in this manual. To support implementation of the model, consider the following steps (click hyperlinked text to go to that section of the manual):

1. At least three times per year, select a CBM task based on the grade level of interest or select all tasks.
2. Administer the CBM task(s) to your entire class (we recommend administering 2-3 prompts of each task).
3. Score student responses (determine the mean of 2 or median of 3 scored prompts).
4. Determine the class mean by taking the average score.
5. Compare the class mean to your school or district benchmarks, if available, or to the benchmarks provided in the CBM Toolkit.
6. If the entire class is performing below benchmark, identify a research-based writing program to put into place for the entire class.
7. If most of the class is performing at or above the benchmark, identify those students who are performing below benchmark and graph their CBM data.
8. Choose high quality interventions for those students by using the *Instructional Alignment Tool* and *Diagnostic Checklists* (in the Writing Intervention Toolkit) to determine an appropriate area of focus.
9. Group your students based on similar needs.
10. Create a Writing Instructional Plan (WIP see templates in the Writing Intervention Toolkit) for each group.
11. Implement high quality instruction with fidelity for 6-8 weeks using the WIP for each group.
12. Monitor student progress during the intervention with a different CBM prompt (using the same type of task) each week.
13. Review each student's progress monitoring data at the end of the intervention period. Use decision-making rules to determine if instructional changes are necessary.

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Appendix A: Sample DBI Forms

Mrs. Lewis's *Selecting an Appropriate CBM Task* form for Molly

Mrs. Lewis's *Writing Instructional Plan* for Molly

Decision-Making Rubric

Mrs. Lewis's *Decision Log* for Molly

Selecting an Appropriate CBM Task

1. Identify a student who is struggling in writing: Molly

2. Identify this student's level of writing performance:

   For each, place a check mark in the box indicating the area(s) in which the student is struggling.

Transcription	**Handwriting**	**Spelling**	**Mechanics**
		X	X
Text Generation	**Words**	**Sentence**	**Passage**
		X	X

3. Based on the above information, identify the area that is most problematic for this student.

Student is…	***struggling with transcription***	***struggling with text generation***
developing word-level skills	Use Word Dictation	
developing sentence-level skills	Use Picture-Word	Use Picture-Word
developing passage-level skills		Use Story Prompt

**Also consider: What is the goal for this student by the end of this school year?

Writing sentences with correct spelling and mechanics as Molly must know how to write sentences before she can be expected to write a paragraph. Molly knows all letter names and can clearly form each letter correctly.

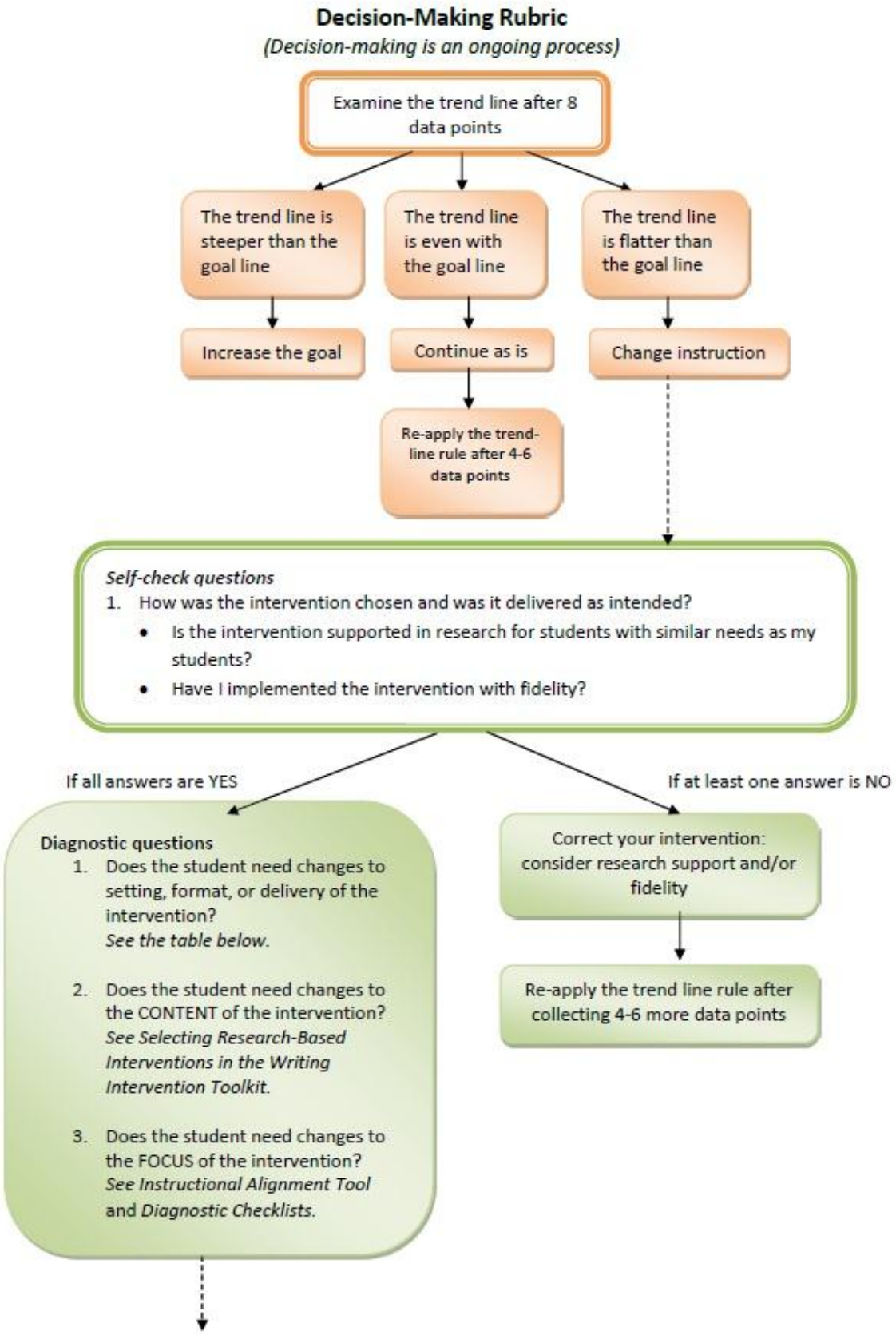
4. Determine the CBM task and scoring method for monitoring this student's progress based on his/her present levels of writing performance.
   - Possible scoring methods to use: WW, WSC, CLS or CWS, ILS or IWS.
   - For students who are struggling with word-level writing skills, scoring CLS on Word Dictation may be the most sensitive measure.
     - Consider that CWS is a more comprehensive measure of writing skill development than WW or WSC. It is also associated with higher alternate form reliability and criterion validity.

Picture-Word Prompt and score using Correct Word Sequences (CWS).

5. Using the measure identified in #3 above, administer three ***different*** prompts of the ***same task*** within one week and take the median score to establish baseline.

Writing Instructional Plan (WIP) Molly Example

Activity	Time	Materials
Each session: Review Objectives and Rules ☐ Introduce/review the purpose of the lesson and objectives ☐ Introduce/review rules and your own motivational system	<1min	☐ Rules ☐ [insert your own motivational materials].
Warm-Up	2 min	
SPELLING ***Days 1, 3, & 5: Word Building Mini-Lesson*** ☐ Day 1: Introduce rime (e.g., an), build words on cards & worksheets ☐ Day 3**:** Introduce new rime (e.g., ad), build words on cards & worksheets ☐ Day 5: Review both rimes ***Days 2 & 4: Word Study Mini-Lesson*** ☐ Students study how to spell words they misspelled from possible content sources (e.g., writing samples) ☐ Once a word is mastered, it is removed and a new word is added.	7 min	***Days 1, 3, & 5*** ☐ Word-building cards ☐ Worksheets ***Days 2 & 4*** List of words ☐ Word Study Chart ☐ Paper & pencil ☐ Word Study Cards
SENTENCE WRITING ***Days 1-5: Sentence Construction Mini-Lesson*** ☐ Focus on capitalization and punctuation	10 min	Practice Sheets (from Mini-Lesson TG1.1)
Days 1-5: **INTEGRATED WRITING PRACTICE** ☐ Review words covered in previous activities and show how to make sentences using these words using capitals and punctuation.	5 min	*Writing Mini-Lesson*
Each session: Closing Lesson wrap-up: summarize words that were practiced and rules for capitalization/punctuation.	<1 min	
Once per week: CBM-W administration ☐ Administer a CBM prompt to students ☐ Score and graph as soon as possible following the lesson	3 min	CBM-W probes, Pencils, Timer


Decision-Making Rubric
(Decision-making is an ongoing process)
Examine the trend line after 8 data points
The trend line is steeper than the goal line
The trend line is even with the goal line
The trend line is flatter than the goal line
Increase the goal
Continue as is
Change instruction
Re-apply the trend-line rule after 4-6 data points
Self-check questions
1. How was the intervention chosen and was it delivered as intended?
• Is the intervention supported in research for students with similar needs as my students?
• Have I implemented the intervention with fidelity?
If all answers are YES
If at least one answer is NO
Diagnostic questions
1. Does the student need changes to setting, format, or delivery of the intervention? See the table below.
2. Does the student need changes to the CONTENT of the intervention? See Selecting Research-Based Interventions in the Writing Intervention Toolkit.
3. Does the student need changes to the FOCUS of the intervention? See Instructional Alignment Tool and Diagnostic Checklists.
Correct your intervention: consider research support and/or fidelity
Re-apply the trend line rule after collecting 4-6 more data points

Directions for Choosing Instructional Changes

1. *Ask the following questions and choose **ONE** option to change your intervention.*
   If you are not sure which question fits your student best, start at the far left column of Changes to Setting and Format or Changes to Delivery, and move from left to right.

Changes to Setting and Format			Changes to Delivery		
Does the student need more time in instruction?	Does the student have a problem with attention or motivation?	Does the student need more individualized interaction?	Does the student need more opportunities to practice?	Does the student need more explicit instruction?	Does the student need more systematic instruction?
↓	↓	↓	↓	↓	↓
Reteach the skill Increase the length of the instruction	Shorten the activity segment Take more frequent breaks With student input, identify ways to increase on-task behavior through self-monitoring strategies Have the student complete a motivation inventory Provide a real world context for the skill Give the student choices related to activity options Have student self-monitor progress and set his/her own goal Teach student to use positive self-statements or reframing	Spend more one-on-one time during small group intervention Spend more one-on-one time outside small group intervention Add peer-mediated component to existing activities	Offer individual practice opportunities to the student Check frequently for student understanding using various methods of response Provide specific feedback that relates to student goals and completion of tasks	State the purpose and learning goal of each lesson, along with rationale Provide models with clear explanations Provide guided practice opportunities Use pictures, manipulatives, and "think-alouds" Repeat the directions Provide immediate and specific feedback	Reteach foundational skills that might have been missed Break down tasks into smaller steps Break down instruction into simpler segments Use step-by-step strategies Provide scaffolded support that can be reduced over time

2. *Implement your change, document on the Change in Intervention Plan (CIP) form, continue monitoring progress, and repeat the entire decision-making process to evaluate the effects of your instruction.*

Edit this fo

DBI Decision Log

Please use this form to document any changes to your writing intervention

* Required

Teacher ID

Please enter the 4-digit PIN assigned to you at the beginning of the study.

1111

Student Initials *

Please enter the first and last initials of your target student.

MO

Date of Instructional Decision *

Please enter the date of the instructional change. Be sure to indicate this decision by inserting a vertical line on your graph.

11/01/2015

Please indicate the TYPE OF DECISION you made. *

- ○ Keep instruction as-is for now.
- ○ Raise the student's long-term goal.
- ◉ Change instruction.
- ○ Other:

Please indicate the REASON for your decision. *

- ○ The student is on track.
- ○ The student is making more progress than expected.
- ◉ The student is not making sufficient progress.
- ○ Other:

If you decided to change instruction, please indicate what TYPE of change(s) made.

Check all that apply. See Decision-Making Rubric for specific options.

- ☐ Change in writing focus (e.g., handwriting, spelling, sentence-writing skills, passage writing skills, planning, organization, etc.)
- ☐ Change in content (sounds, letters, words, vocabulary, genre, etc.)
- ☑ Change in time
- ☐ Change in motivational strategies
- ☐ Change in group size

- [ ] More practice
- [ ] More explicit instruction
- [ ] More systematic instruction
- [ ] Other:

Please describe your change in detail (for example, "I decreased groups size from 5 to 3"). *

Increased instructional days from 3 to 5 days per week.

Submit

100%: You made it.

Never submit passwords through Google Forms.

Appendix B: Research on CBM for Beginning Writers

Recent research has demonstrated each of these tasks has sufficient reliability and criterion validity. Recent research also shows each of these tasks produces reliable slopes that are sensitive to growth when the tasks are administered for at least 8 weeks. These findings are summarized in Table B1.

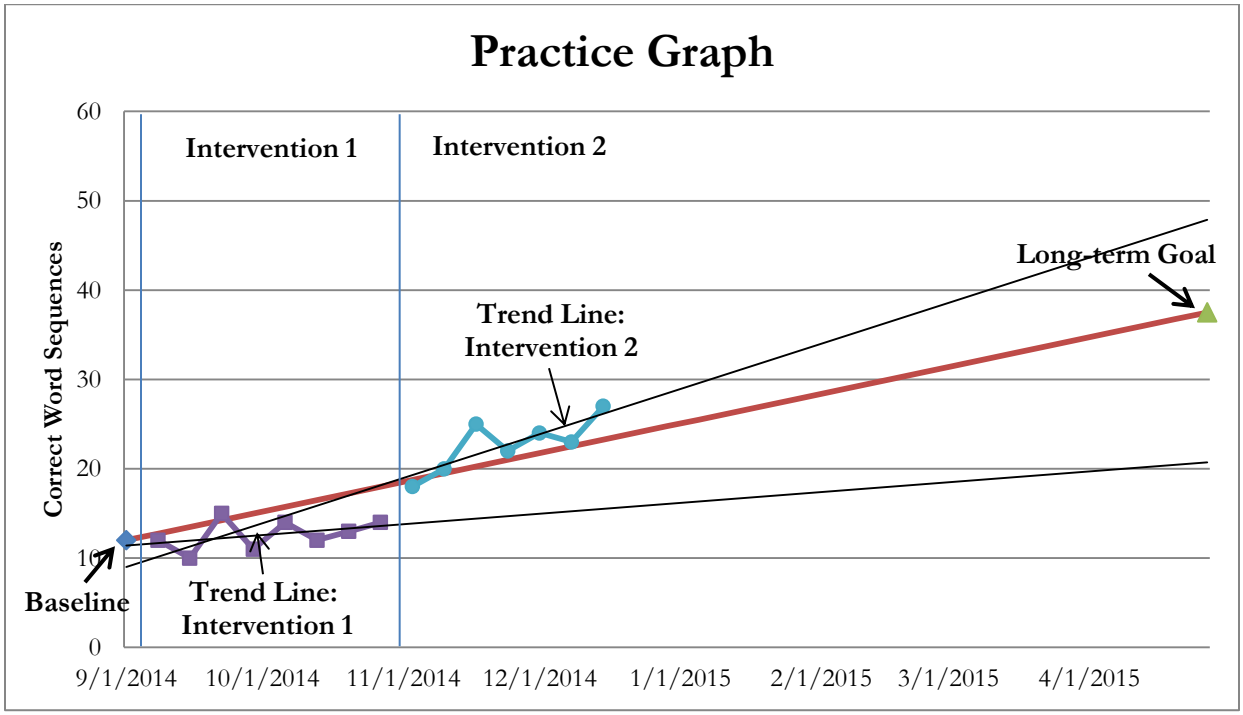
Table B1. ***Reliability, Validity, and Sensitivity to Growth of CBM Tasks.***

CBM Task/Score (Grade)	Reference	Reliability	Criterion Validity	Sensitive to Growth
Word Dictation/CLS (Grades 1-2)	Hampton et al. (2010), Lembke et al. (2003)	r >.70	r = .51-.61 (Criterion: TEWL)	Yes
Picture-Word/CWS (Grade 1)	McMaster et al. (2009, 2011)	r >.70	r = .50-.60 (Criterion: TOWL)	Yes
Story Prompts/CWS (Grades 1-3)	McMaster et al. (2009, 2011, 2012)	r >.70	r = .50-.65 (Criterion: TOWL)	Yes

Note. CLS = Correct Letter Sequences, CWS = Correct Word Sequences.

Appendix C: Practice Data and Graph

Date	Baseline	Goal Line	Long-Term Goal	Intervention 1	Intervention 2
9/1/2014	12	12			
9/8/2014		12.75		12	
9/15/2014		13.5		10	
9/22/2014		14.25		15	
9/29/2014		15		11	
10/6/2014		15.75		14	
10/13/2014		16.5		12	
10/20/2014		17.25		13	
10/27/2014		18		14	
11/3/2014		18.75			18
11/10/2014		19.5			20
11/17/2014		20.25			25
11/24/2014		21			22
12/1/2014		21.75			24
12/8/2014		22.5			23
12/15/2014		23.25			27
12/22/2014		24			
12/29/2014		24.75			
1/5/2015		25.5			
1/12/2015		26.25			
1/19/2015		27			
1/26/2015		27.75			
2/2/2015		28.5			
2/9/2015		29.25			
2/16/2015		30			
2/23/2015		30.75			
3/2/2015		31.5			
3/9/2015		32.25			
3/16/2015		33			
3/23/2015		33.75			
3/30/2015		34.5			
4/6/2015		35.25			
4/13/2015		36			
4/20/2015		36.75			
4/27/2015		37.5	37.5		


Practice Graph
Correct Word Sequences
60
50
40
30
20
10
0
Intervention 1
Intervention 2
Long-term Goal
Trend Line:
Intervention 2
Baseline
Trend Line:
Intervention 1
9/1/2014
10/1/2014
11/1/2014
12/1/2014
1/1/2015
2/1/2015
3/1/2015
4/1/2015